THE MAGAZINE OF THE ACTUARIES INSTITUTE
MARCH 2012 ISSUE 167

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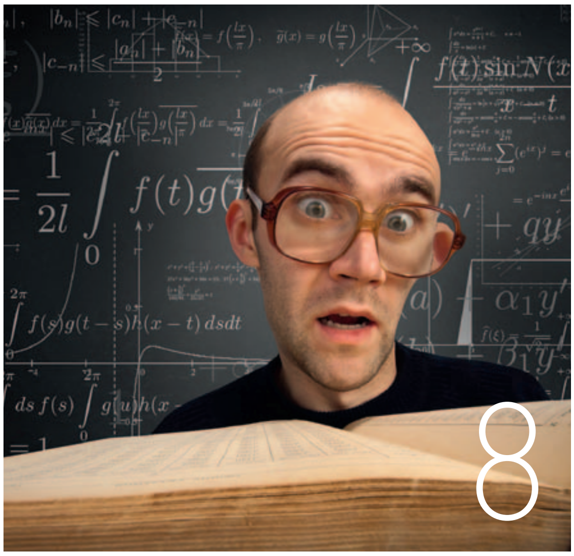

Actuaries

CONTRIBUTIONS
Contributions should be sent to the Actuaries Institute, marked to the attention of Katrina McFadyen (Communications and Marketing Manager) at:
katrina.mcfadyen@actuaries.asn.au
All contributions must conform to our submission guidelines which are available from the Communications and Marketing Manager.

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Coming up

March

Insights
The New DB Funding Framework
Tuesday 27 March 2012, Sydney

The Superannuation Practice Committee will share their views on the proposed Prudential Standards in relation to DB funding.

Young Actuaries Program
Part III Exams – the quest for the Holy Grail
Wednesday 28 March 2012, Melbourne
Experienced past exam markers share their tips and knowledge on achieving success in the Part III Exams.

Insights
The New DB Funding Framework
Thursday 29 March 2012, Melbourne

April / May

Leadership Forum
The Who, What, When and How of Leadership
Wednesday, 4 April 2012, Sydney

Develop your leadership skills and become part of a network of members with an interest in leadership.

Young Actuaries Program
Part III Exams – the quest for the Holy Grail
Wednesday 11 April 2012, Sydney

Insights Networking
Basel II Retail Modelling Approaches
Wednesday, 18 April 2012, Melbourne

As debate rages over the strength of Australian house prices and more generally, the economy, the ability to communicate a robust framework for the modelling of retail default rates is more vital than ever.

Insights
New Insurance Capital Standards – the Impact on the Role of the Appointed Actuary
Wednesday, 18 April 2012, Sydney

The Life Insurance and Wealth Management Practice Committee consider the role of the Appointed Actuary under the proposed new capital standards.

Financial Services Forum 2012
Think Big
Monday 30 April – Tuesday 1 May 2012 , Melbourne

Register by Friday 6 April to take advantage of the early bird discount. Don't miss out as seats are filling fast.

July / August

Fellowship and Presidential Dinner
Tuesday 17 July 2012, Melbourne

President David Goodsall invites members and their guests to join him in welcoming new Fellows and Actuaries.

Fellowship and Presidential Dinner
Tuesday 14 August 2012, Sydney

Editorial

Ben Ooi
editor@actuaries.asn.au

Hello and welcome to the first edition of *Actuaries* magazine: a new name, a new look and most importantly, a renewed focus to increase reader engagement.

At this point, I should introduce myself as the new Editor for the magazine. Having just finished an eight-year stint authoring the Institute's Private Health Insurance newsletter, I wanted to try a slightly longer publication (and I thought to myself "what could possibly go wrong?"). So bring on the adventure!

What better way to start off a new year than with a refresh and given the relatively new corporate logo and name, it seemed an opportune time to bring consistency into our corporate branding – hence the magazine's name change, which now matches the corporate logo and the Institute's web portal URL.

As part of the review, I am delighted to announce Andrew Brown's new regular column "*Actuaries taking the lead*" which will aim to give actuaries insights into leadership, communications and soft skills, and effectively assist us to be more than actuaries!

Over the next few months, based on your feedback, the magazine will evolve further to incorporate:

- more human interest pieces from you, where readers can write in about activities outside the office such as wider community activities;
- an expanded *Actuary Unearthed* (renamed *Under the Spotlight*) which has increased its scope to capture profiles of not just individuals, but also groups or teams;
- a new photography competition where readers can submit entries for the front cover(s) subject to specified requirements;
- more profiling articles on inspiring actuaries; and
- an attempt at being more iPad friendly (as well as other e-readers and tablets; we will try what we can!)

Some say change is accompanied by risks and challenges. In my humble opinion, change can also often be fun and exciting. Keeping with the spirit of my interview with David Goodsall whose key message is to have a more adventurous spirit and "have a go", I will be experimenting with the magazine throughout the year! I know some of you may say "if it ain't broken, why fix it?" I think the recent collapse of Kodak illustrates that we have to dynamically move with the times to stay relevant. I hope that you will find these changes positive and invite you to write in and tell us what you think.

Finally, I wish to thank Catherine Robertson-Hodder and James Collier for their tireless efforts as editors for the past two years. I would also like to welcome Solai Valliappan who joins the Editorial Committee as an Assisting Editor. A

A conversation with David Goodsall

Take a punt! Stick your neck out! Be engaged! There were the messages that stuck in my head as I left my interview with David Millington Goodsall, Director, former Senior Partner of a consulting firm, thinker, leader, an accomplished man with a long career in financial services, and now President of the Actuaries Institute.

Why did you take up the presidency?
I've got a lot out of being an actuary and its important to give something back. I've been involved with the profession for a long time, through education as a tutor and marker and then on various committees plus two terms on Council. As President, I hope to bring some of the experience and business skills I have developed over the years to the profession to realise our strategy, so that we can increase our sphere of influence and attract the best professionals of the future.

What do you think are the main issues for the profession at the moment and the future?
We are facing a lot of change in the world and the traditional practice areas are no longer the obvious employer of new actuaries and we need to retain people in the profession after they finish their education. Unfortunately actuaries are often perceived as technical experts rather than business advisers with powerful technical skills. Globally, even where the traditional areas are much larger and still growing, actuaries are facing this problem, so really we have to make sure that actuaries are not seen as just technical people and that they are appropriately involved in the commercial side of the businesses. We are at risk of becoming much less relevant than we are today.

If you look at how many actuaries are on the C suite of insurance companies today compared to thirty years ago it's very telling.

What would you like to have achieved during your term and what legacy do you hope to pass on as president?
I hope to spark debate on issues such as

- Are we attracting the right people into the profession?
- Do we know how to take risk? Because if you are in risk management and avoid risk I think by definition you are not a good risk manager.
- Where are we going to be in ten years? Who is going to be employing actuaries then and how do we equip the profession to meet their needs?

I would like to see an active debate about the future actuary and what action we need to take, so that future leaders of the profession have a clear roadmap supported by our members. I hope that people will become involved because it's the Members who will create change.

What is your personal development goal by being president this year?
I'm always learning. Through interaction with the membership, I hope to learn from the younger members what they see being a profession in the 21st century is all about, so I am really keen to get input from them.

What is your main challenge?
A real challenge is getting the input and engagement from the membership who are often time poor and inundated with data. We are trying a number of social media and electronic means as well as physical meetings to reach the members but they are only effective if people use them.

I have had a number of comments on my blog so far but they are generally from people of my generation. It's the younger people who are going to be leading the profession in the future so we need to know what they think.

How do you attract the right people?
I think it comes down to making the education attractive for people with the right skills when they are making a tertiary education choice. We, like all the other bodies around the world, have a very technical education and I think we need to build more commercial skills into the education.

We have to demonstrate a clear connection between the profession and an interesting career, which used to be there. I have had a really interesting career and it is what you make of it but we need people to stand up and show how interesting it is and really promote it. This begins in the schools and universities, so if you have a chance to speak to a school careers night, do it.

So how do you encourage a more adventurous spirit?
I started uni not long after the actuarial course at Macquarie University came into being. Before Macquarie University the usual way to become an Actuary was to do a university degree like maths or science, then you put that aside and started again, doing all the A subjects and the Fellowship and built another career. That was a big risk. So to get into the profession you effectively had to be a risk taker.

We need to capture more of that type of person and keep those we already attract. I don't have the solution yet and I'm open to ideas.

Is there something that the Actuaries Institute can do to boost the image that actuaries are more than technical people?
The Society of Actuaries has spent a lot of money on doing exactly that and they have done a very good job but only scratched the surface. It takes time and resources.

We're a small organisation so what we can do is limited and we achieve a lot for our size. We do a lot in the public policy area and this has raised the public profile but the image of actuaries with employers comes

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from our members, because that's what they see. The employers see the activities of our members so if they see actuaries rising to the occasion and adding value that will change their perception.

What about role models to inspire young actuaries?
There are many good role models out there and we need more to share experiences and guide younger members. I hope they will take the opportunities offered. It can be a bit hard, our mentor programme is no longer being run because we couldn't get the members to take it up. Quite a few of those who did participate didn't really take advantage of it.

So where is the profession going to be in ten years time?
We're currently a mainstream profession and well regarded by society. I think it could slip back into a statutory compliance role. That would be unfortunate. I would hope that we can evolve to the point where an actuary is seen as a business focussed professional adding value through a strong analytical approach.

Do you think of the new breed of actuaries are too specialised or should they be more generalists?
I came through the old system where I did life, general, super and investments. Understanding the different approaches taught me there are many ways of looking at a problem and you need to look for the best technique or approach. Today I think we should go beyond traditional areas and teach broader skills. So areas such as risk management and the commercial aspects of data analytics are relevant to the business world where we can apply our analytical approach combined with complex judgement. Because that's something that really sets us apart, we are one of the few professional bodies that train people in complex judgement.

How would you explain what an actuary does?
I like to say actuaries are skilled at using a variety of analytical approaches, often involving probability, to take usually incomplete information from a number of sources and come up with a practical solution.

Should we actuaries have to keep explaining what we do? Do we need TV ads?
We don't have the money for that, but we get a lot of value from actuaries who demonstrate good business and professional skills. one way of doing that is If someone asks for your opinion, give it and give it quickly. Don't hedge your bets.

What qualities do you think a good actuary should have?

- A strong analytical mind and skills.
- Know your subject matter.
- Be a lateral thinker.
- Think of things from the other person's point of view, add value and communicate effectively.
- Be aware that you always have a lot to learn.

Where will you be in five years time?
I'm really enjoying my work now which is quite varied. I'm meeting many interesting people and learning so much all the time which is sparking new ideas so I hope a couple of these will work out and I'll still be enjoying work, perhaps with a little more time after being President.

What is a general message you want to pass on to readers?
Being an actuary means being part of a wonderful profession that gives you the platform to have a really interesting career if you want to take advantage of it. Participate in the profession as much as you can. In your career, look for opportunities and take them rather than wait for them to find you. A

I like to say actuaries are skilled at using a variety of analytical approaches, often involving probability, to take usually incomplete information from a number of sources and come up with a practical solution.

in focus: Lesley Traverso

Lesley's Recent Placement:

Client:	Deloitte Actuaries & Consultants, Sydney
Position Filled:	Partner
Successful Candidate:	Rick Shaw

Rick's background
I have been working overseas for many years and for most of the last ten years Bermuda; first, as the country leader for Tillinghast, and then as Chief Actuary for the Bermuda Monetary Authority (BMA). I've worked on many interesting projects including leading a large team estimating World Trade Centre losses post 9/11 and reviewing the southern U.S. hurricane losses, including Katrina.

How did you become aware of the role at Deloitte?
Lesley Traverso approached me at the Institute's General Insurance Seminar in November 2010 to talk about the role. As I am not normally a natural fit with office culture, I was inclined to say 'thanks, but no thanks', however, Lesley is quite persistent and ultimately I decided to listen! I soon realised this wasn't going to be a traditional general insurance office job; Deloitte were looking for someone to head up and build a substantial GI practice, working closely with its Strategy, Operations, Technology and Data Analytics practices. This was very attractive to me as it involved working with clients, to help build their strategic vision and to make that vision a reality.

How is it going?
I love a challenge. Over the years I have built practices in a number of countries, but not at home in Australia. It's still early days and I am a little overwhelmed by how much I have to learn, but I am starting to sense the enormous opportunities in the market.

What are your plans for the practice?
I want to build the team. We are looking for outstanding actuaries - actuaries who are technically at the top of their game, and who are creative. The actuaries I am looking to hire need to be innovative, committed to the pursuit of excellence and with a burning desire to become thought leaders.

How would you describe your experience with Lesley and DW Simpson?
I have not been that impressed by recruiters before this experience, either as an employer or when looking for work. Lesley got me over the line on this, and I owe her a big debt. I gave her some feedback about my past experiences and Lesley really listened and responded. In my mind, Lesley is one of the premier recruiters in the actuarial field in Australia. We're a club, and Lesley's an honorary member. DW Simpson's international network opens up the world for Aussie actuaries and I think this is great.

Profile

Title: Managing Partner - International, Executive Vice President, DWS Global.

Role Description: Leading multiple teams in offices across all continents outside of the Americas and Bermuda. Recruiting at a senior level. Keynote Speaker at conferences across the globe. Active and advisory member on a number of committees. Board member of the DW Simpson Global Leadership team.

Strengths: Professional. Tenacious. Seeing beyond the job description. Matching people to roles they wouldn't think of. "Oh and dancing on tables at conferences!"

Weaknesses: "Chocolate. Chocolate. Chocolate....did I mention chocolate?"

Hobbies: "Flying flimsy planes, very high in the sky and doing scary twirly diving manoeuvres!"

Contact Details:

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lesley.traverso@dwsimpson.com

Current Assignment:

Focus on Re-insurance! Focus on Singapore!

Singapore is a hive of actuarial re-insurance activity at the moment, with a number of our key clients looking to expand the size of their teams by recruiting Actuaries into interesting life and non-life positions. Australian actuarial experience is well respected in Singapore and the market there can offer Actuaries an exciting career. There are a variety of career options, including pricing, marketing, capital management and valuations. The level of experience can range from just three years upwards, with many organisations also looking to deepen bench strength at the senior level. Asia offers exciting opportunities for the go-getter and Singapore is fast becoming a regional hub. With all round warm weather, low tax rates and a dynamic business environment which is rapidly developing, Singapore offers an enticing place for entrepreneurial professionals to thrive.

Call me today to discuss opportunities in Singapore as well as other roles overseas and in Australia.

Solai Valliappan
solaiv@gmail.com

Solai Valliappan

Name... Solai Valliappan

Title... Actuarial Analyst (GI)

Organisation... Zurich

Summarise yourself in one sentence... "The glass is always half full"

My interesting/quirky hobbies... TV series marathons. I've watched Boston Legal and The Thick of it, now for West Wing

My favourite energetic pursuit... Salsa dancing

The sport I most like to watch... Tennis, but tipping season is starting soon so in winter AFL

The last book I read (and when)... I'm anticipating that I'll finish Stephen Fry's second autobiography 'The Fry Chronicles' by the time this goes to the printers

My favourite artist/album/film... Artist – Phoenix; Album – Hopes and Fears by Keane; Film – You've got mail

The person I'd most like to cook for... My grandmother because she always spoils me so much with her cooking

I'm most passionate about... Chocolate cake. Family and friends also rate highly, and I think it's important to give back to charity with your time

What gets my goat... Arrogance and slow walkers (especially at lunch time!)

I'd like to be brave enough to... Scuba dive

In my life I'm planning to change... The number of countries I have travelled to

Not many people know this but I... am a new Assistant Editor on the Actuaries magazine Editorial Committee

Four words that sum me up... Friendly, trustworthy, astute, sociable

What I wanted to be when I grew up... An artist (I was 7)

Why and how I became an actuary... The job guide said the course would be 80% male... sadly that statistic was not reflective of my cohort. On a more serious note, the mix of business and maths was quite appealing and the offer of a scholarship sealed the deal

Where I studied to become an actuary and qualifications obtained... Bachelor of Commerce at UNSW on the actuarial studies co-op program (Part 1 and 2). I'm an Associate currently half way through Part 3 exams

My work history... My 3 co-op placements during uni were at Finity, APRA and Suncorp. I started at Zurich as a graduate, spent a few years in pricing and now about a year in the valuation team

What I find most interesting about my current role... Interacting with other parts of the business – underwriting, claims, pricing actuaries, accounting, reinsurance and senior management. I also enjoy how my work leads to interaction with people both in Australia and overseas – Switzerland, Hong Kong, London and even Bermuda!

My role's greatest challenges... Working with different time zones. Because of where we're placed in the world by the time you read an overseas email you've already lost 13 hours before the deadline. There's also the occasional Friday night phone call with Switzerland that clashes with drinks

Who has been the biggest influence on my career (and why)... Robert Thomson (APRA) – how to explain complicated concepts, James Makin (Zurich) – best mentor ever!, David Whittle (Zurich) – insightful knowledge and Elsa Cheung (Zurich) – peer support

My proudest career achievement to date is... This may sound trivial to seniors, but I'm sure my peers will agree that securing a graduate position at the height of the GFC was certainly a big deal

10 years from now, I will be... X+10 years old

When I retire, my legacy will be... Promoting the profession in a positive light

The most valuable skill an actuary can possess is... Confidence

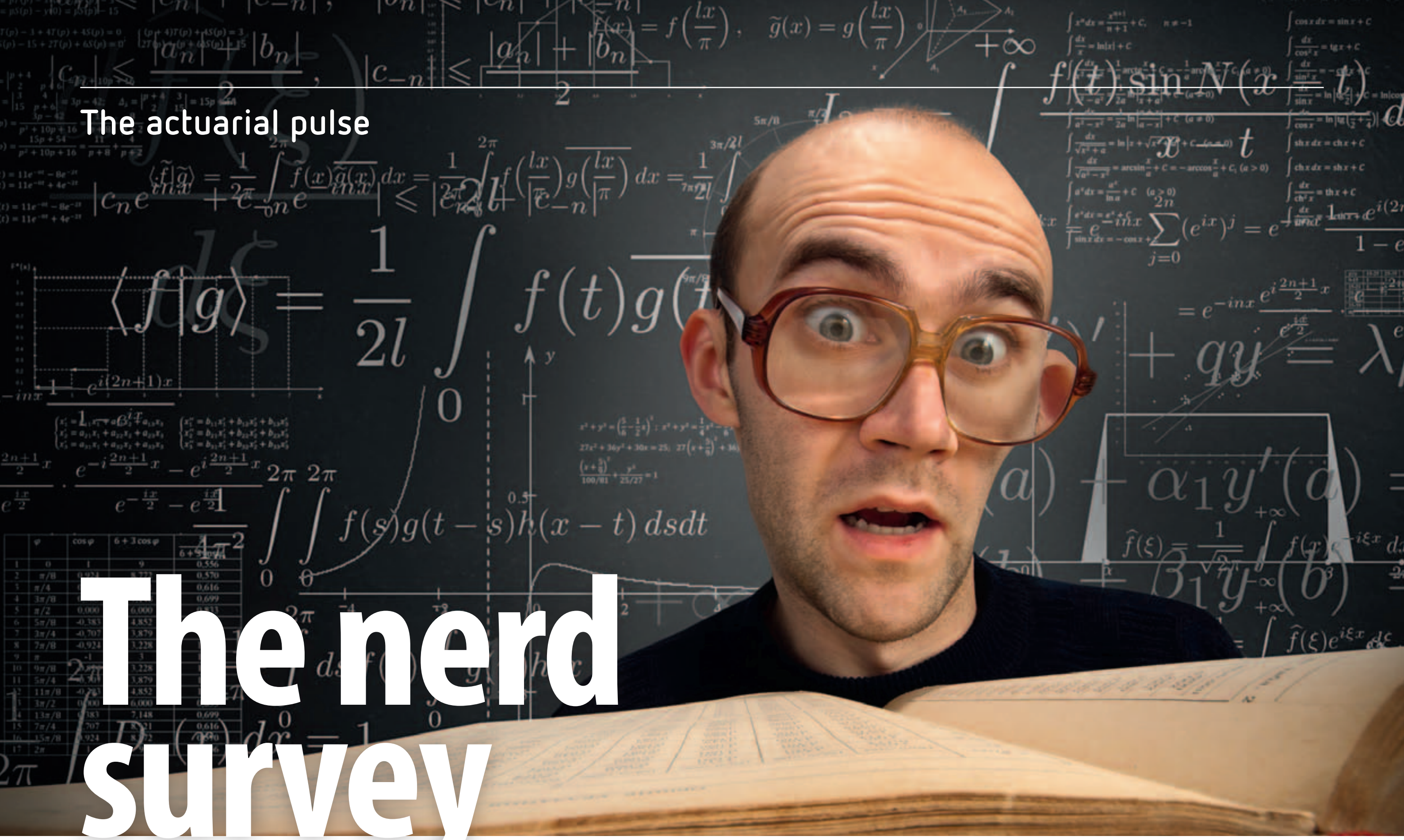
If I was President of the Institute, one thing I would improve is... Get a twitter account – see the President's blog "To tweet or not to tweet? That is the question"

At least once in their life, every actuary should... Underwrite some business

My best advice for younger actuaries... Seek a mentor

If I could travel back in time I would... See the dinosaurs

If I win the lottery, I would... Buy an Aston Martin (the DB9)



The nerd survey

NEW SURVEY QUESTIONS WILL BE AVAILABLE IN APRIL 2012. WHAT WOULD YOU LIKE TO KNOW? IF YOU HAVE A QUESTION YOU WOULD LIKE TO PUT TO THE MEMBERSHIP, EMAIL IT TO EDITOR@ACTUARIES.ASN.AU

RESULTS: REPORT GENERATED ON 21 FEBRUARY 2012. 417 RESPONSES.

Once upon a time, being a nerd was considered to be a bad thing. Typically defined as people who are intelligent but socially inept, with an intense interest in subjects that others tend to find either difficult to comprehend (such as science, maths and technology) or silly and childish (such as comics and video games), nerds were seen to be objects of ridicule, as exemplified by the nerd characters in movies of the eighties such as *The Breakfast Club* and *Sixteen Candles*.

However, with the rise in importance of maths and technology over the last few decades, there has also been an associated rise in the status of nerds. Nerds have gone from zeroes to heroes and now everyone wants to get in on the action. In 2011, a movie about nerds (*The Social Network*) was nominated for the Best Picture Oscar; a Roy Morgan survey found that a show about nerds (*The Big Bang Theory*) was Australia's most loved TV program; a trilogy of novels about a nerd (Stieg Larsson's *Millenium* books) appeared in the bestseller lists; and the death of nerd icon Steve Jobs was seen as a great tragedy for nerds and non-nerds alike.

In this month's *Actuarial Pulse*, we examine the nerd phenomenon by looking at just how nerdy actuaries consider themselves to be, and determining the number of stereotypical nerd traits possessed by actuaries, because, after all, being a nerd has now become the cool thing to be.

SECTION 1: DEMOGRAPHIC INFORMATION

QUESTION 1: WHAT IS YOUR GENDER?

Female 31% (131) | Male 69% (286)

QUESTION 2: WHAT IS YOUR AGE?

Choice	Count	%
Under 25	39	9%
25 – 34	181	44%
35 – 44	86	21%
45 – 54	64	15%
55 – 64	27	7%
Over 65	18	4%

SECTION 2: HIP TO BE SQUARE?

QUESTION 3: IN GENERAL, DO YOU CONSIDER ACTUARIES TO BE NERDS?

Yes 68% (283) | No 32% (131)

QUESTION 4: DO YOU CURRENTLY CONSIDER YOURSELF TO BE A NERD?

Yes 39% (163) | No 61% (254)

Are actuaries nerds? According to respondents, the answer to this question appears to be Yes and No. 68% of respondents think that actuaries are nerds, but only 39% consider themselves to be nerds, with 35% of respondents answering 'yes' to both questions. However, as can be seen from the graph below, the proportion of 'yes' responses to these questions varies with age.

Respondents in the younger age groups are more likely to answer 'yes' to both of these questions. There are many possible reasons for this, but one theory is that this is a reflection of the changing

Genevieve Hayes
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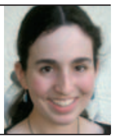
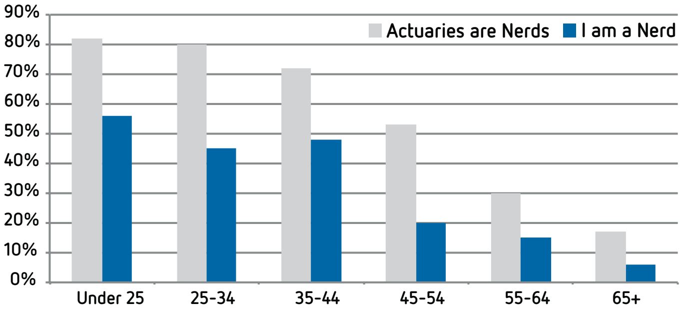
The Actuarial Pulse is an anonymous, web-based survey of Institute members, run on a monthly basis, giving members an opportunity to express their opinions on a mixture of serious and not-so-serious issues.





attitudes towards nerds described previously. Older respondents, who came of age at a time when nerds were shunned, might not want to identify themselves and their profession with nerdishness. Younger respondents, some of whom would have only ever known a nerd-friendly world, are less likely to have such concerns.

A higher proportion of females than males consider actuaries in general to be nerds (77% vs 64%), but approximately the same proportion of females and males consider themselves to be nerds (40% and 38% respectively).

QUESTION 5: ARE YOU PROUD OF YOUR ANSWER TO QUESTION 4

Yes 80% (329) | No 20% (82)

Turning our attention to whether actuaries consider their nerdishness to be a good thing or a bad thing, it is pleasing to see that the majority of respondents are proud of their answer to Question 4, including 87% of those who identified themselves as nerds (as opposed to only 74% of non-nerds). Nerd pride is high in all age and gender groups, with the highest proportions of proud nerds being in the 35 – 44, and over 55 age bands (95% and 100% respectively), and among females (94%).

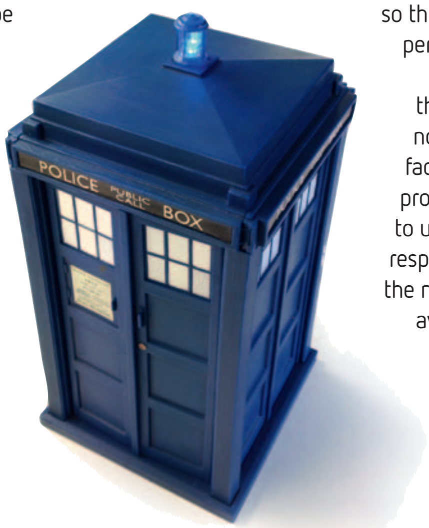
QUESTION 6: WERE YOU CONSIDERED TO BE A NERD IN HIGH SCHOOL?

Yes 55% (228) | No 45% (187)

The purpose of this question is to test the theory that actuaries are likely to have been considered to be nerds in high school, due to the fact that nerds and actuaries both tend to be good at math. The responses to this question are split almost exactly fifty-fifty, which on the surface seems a very low proportion of 'yes' answers. However, if compared to the population in general, it would probably be relatively high.

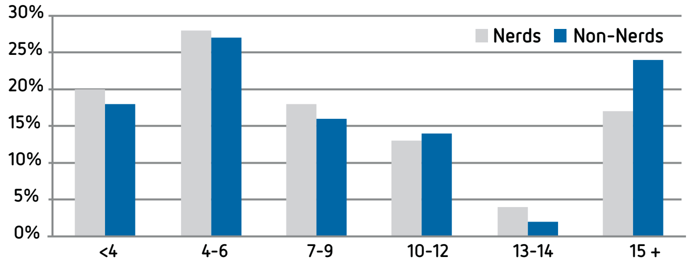
Once again, younger respondents are more likely to have been considered to be nerds in high school than older respondents, and unsurprisingly, 81% of those who considered themselves to be nerds in Question 4 were also considered to be nerds in high school.

The remaining questions in this survey consider whether or not respondents possess some of the traits and interests commonly associated with nerds. For the purposes of the following sections, respondents are divided into two groups, nerds and non-nerds, according to their answer to Question 4.

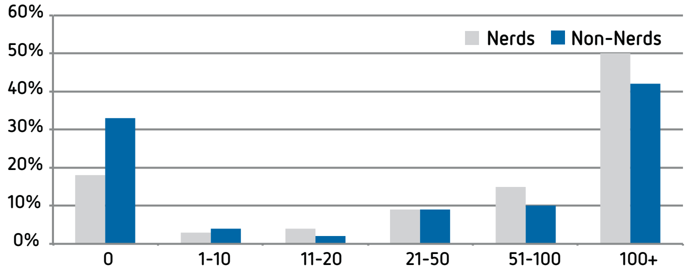


SECTION 3: FRIENDS

QUESTION 7: HOW MANY GOOD (REAL) FRIENDS DO YOU HAVE?



QUESTION 8: HOW MANY FACEBOOK FRIENDS DO YOU HAVE?



In spite of the fact that nerds are stereotypically believed to be socially inept, the distribution of good friends is reasonably similar for nerds and non-nerds. Based on these results, it is estimated that the nerds have 8.05 good friends on average, with the average for non-nerds being only slightly higher, at 8.76. According to Oxford University anthropology professor Robin Dunbar, a person's social world usually consists of "an inner circle of five 'core' people and an additional layer of 10", so the figures observed in this survey seem perfectly normal.

From the results to Question 8, it appears that the nerds have more Facebook friends than the non-nerds. However, this result is distorted by the fact that the non-nerds group contains a greater proportion of older respondents, who are less likely to use Facebook at all. If the analysis is limited to respondents under 35, this situation reverses, with the nerds estimated to have 115 Facebook friends on average and the non-nerds estimated to have 128.

According to Professor Dunbar, people are only capable of managing relations with around 150 friends, so again, these figures seem normal and together with the results to Question 7 form a strong counter-argument to the notion that nerds or actuaries lack social skills.

SECTION 4: MOVIES, BOOKS AND TV SHOWS

QUESTION 9: HOW OFTEN DO YOU QUOTE MOVIES, BOOKS OR TV SHOWS IN CONVERSATION OR EMAILS?

Choice	Nerds (%)	Non-Nerds (%)
Never/Rarely	37%	53%
Sometimes	40%	37%
Frequently	23%	10%

Nerds are notorious for quoting their favourite movies, books or TV shows given the slightest opportunity and this is demonstrated by the responses to Question 9, which show that 63% of nerds sometimes or frequently quote movies, books or TV shows. Nevertheless, quoting is also popular among non-nerds, with 49% of non-nerd respondents indulging in this activity sometimes or frequently.

QUESTION 10: WHAT IS YOUR FAVOURITE MOVIE, BOOK OR TV SHOW QUOTE?

In spite of the fact that quoting is a nerdish past-time, few of the quotes listed in response to this question are from nerdish sources. Admittedly, seven people quoted from each of *The Big Bang Theory*, the *Star Wars* series, and *Lord of the Rings* or *The Hobbit*, all stereotypical nerd-favourites, but the most common quotes given in response to this question were "I'll be back" from *The Terminator* and "You can't handle the truth," from *A Few Good Men*. In addition:

- 20 people quoted *The Simpsons*, including five people who quoted Homer Simpson's catch-cry "D'oh";
- Four people quoted *The Princess Bride*, including two who quoted the line "My name is Inigo Montoya. You killed my father. Prepare to die;" and
- Two people gave "Missed it by that much" from *Get Smart*.

QUESTION 11: WHICH "NERD FAVOURITE" TV/MOVIE SERIES ARE YOU A FAN OF?

Choice	Nerds (%)	Non-Nerds (%)
Buffy/Angel	11%	10%
Dr Who	16%	29%
Lord of the Rings	27%	46%
Monty Python	21%	49%
Stargate	7%	10%
Star Trek	9%	13%
Star Wars	22%	37%
Supernatural	4%	8%
Xena/Hercules	5%	2%
The X-Files	10%	20%

In fact, an examination of the responses to this question reveals that all but two of the "nerd favourite" TV/movie series listed have a greater proportion of fans among the non-nerds than among the nerds, and with the exception of *Lord of the Rings*, *Monty Python* and *Star Wars*, the most mainstream of the series listed, fewer than 30% of either group are fans of any of the other series listed. Only 9% of nerds and 13% of non-nerd respondents claim to be fans of *Star Trek*, often considered to be the quintessential nerd series.

Other nerd-favourite series listed by respondents include *The Big Bang Theory* (mentioned 43 times) and *Battlestar Galactica* (9 times).

SECTION 5: PAST-TIMES AND POSSESSIONS

QUESTION 12: WHICH "NERD ITEMS" DO YOU OWN (AND USE)?

Choice	Nerds (%)	Non-Nerds (%)
Microscope	4%	< 1%
Telescope	6%	6%
Meteorological Equipment	10%	13%
Rubik's Cube (or similar)	28%	21%
Scientific Calculator	68%	54%
Slide Rule	8%	4%
Periodic Table	12%	4%
Video Games Console	46%	24%
Lego/Meccano (or similar)	15%	12%

Like Barbie dolls and action figures, there is a generally held belief that nerds come with accessories, but unlike Barbie, these accessories tend to be more scientific in nature. Nevertheless, the responses to Question 12 show that only relatively small proportions of nerds and non-nerds own any of the items listed. The exceptions to this are Rubik's cubes, scientific calculators and video games consoles, but given that scientific calculators are required for actuarial exams and many people own video games consoles because they also double as DVD/BR players, these two items seem quite ordinary compared to, say, a slide rule. As for Rubik's cubes and other similar puzzles, if these are the nerdiest items that most actuaries own, then there is little to fear among those ashamed of their nerdishness.

Other nerd items owned by respondents to this question include:

- A unicycle;
- A Geiger counter (mentioned by two people);
- Five computers (two people); and
- A pi t-shirt.

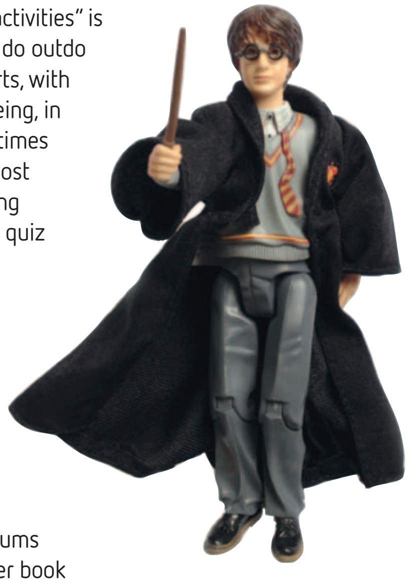
QUESTION 13: WHAT IS YOUR LEVEL OF UNDERSTANDING OF EACH OF THE FOLLOWING NON-TRADITIONAL LANGUAGES?

Language	Never Heard of It (%)	Heard of It (%)	Can Write/ Speak It (%)
Tolkien's Elvish	29%	70%	1%
Klingon	24%	75%	1%
Esperanto	49%	50%	1%
Newspeak (from "*1984*")	56%	40%	4%
Nadsat (from "*A Clockwork Orange*")	81%	18%	1%
Aklo (from the works of H.P. Lovecraft)	96%	4%	0%
Na'vi (from "*Avatar*")	44%	55%	1%
Latin	6%	79%	15%
SAS	15%	50%	35%

Nerds are also commonly believed to be brilliant linguists, specialising in languages that virtually no one actually speaks, but the response to this question suggests that, again, this is a myth rather than reality. Despite the fact that the table above shows only the results to this question for the entire combined sample of nerds and non-nerds, it is still clear that, although many of the languages listed have high name recognition, with the exception of SAS (a computer language used by many actuaries in their jobs) and Latin (which is still taught in some high schools), very few respondents from either group have gone to the trouble of learning to write or speak these languages. Only 14 respondents can actually write/ speak Newspeak (7 nerds and 7 non-nerds), while fewer than five respondents can speak Elvish, Klingon, Esperanto, Nadsat, Aklo or Na'vi.

QUESTION 14: WHICH "NERD ACTIVITIES" HAVE YOU PARTICIPATED IN?

Choice	Nerds (%)	Non-Nerds (%)
Attended a fan convention	10%	1%
Written fan fiction	3%	0%
Sat the Mensa entrance exam	7%	6%
Attended a movie at a midnight screening on opening night	21%	12%
Attended a quiz night	60%	59%
Played Dungeons and Dragons (or similar)	36%	17%
Listed my religion as "Jedi" on the Census	9%	3%
Attended a screening of *The Rocky Horror Picture Show* in costume	5%	2%



Participation in "nerd activities" is the one area where nerds do outdo their non-nerd counterparts, with nerd participation rates being, in many cases, two or more times the non-nerd rates. The most popular nerd activity among those listed is attending a quiz night, with approximately 60% of both nerds and non-nerds having done this, but all other activities in the list are far less popular.

Other "nerd activities" mentioned include • Reading all the interpretive signs in museums • Lining up for Harry Potter book releases • Attempting to solve a $1M maths problem • Becoming an ordained Pastafarian minister in the Church of the Flying Spaghetti Monster.

However, as one respondent commented, "a true nerd doesn't participate. They do their own thing."

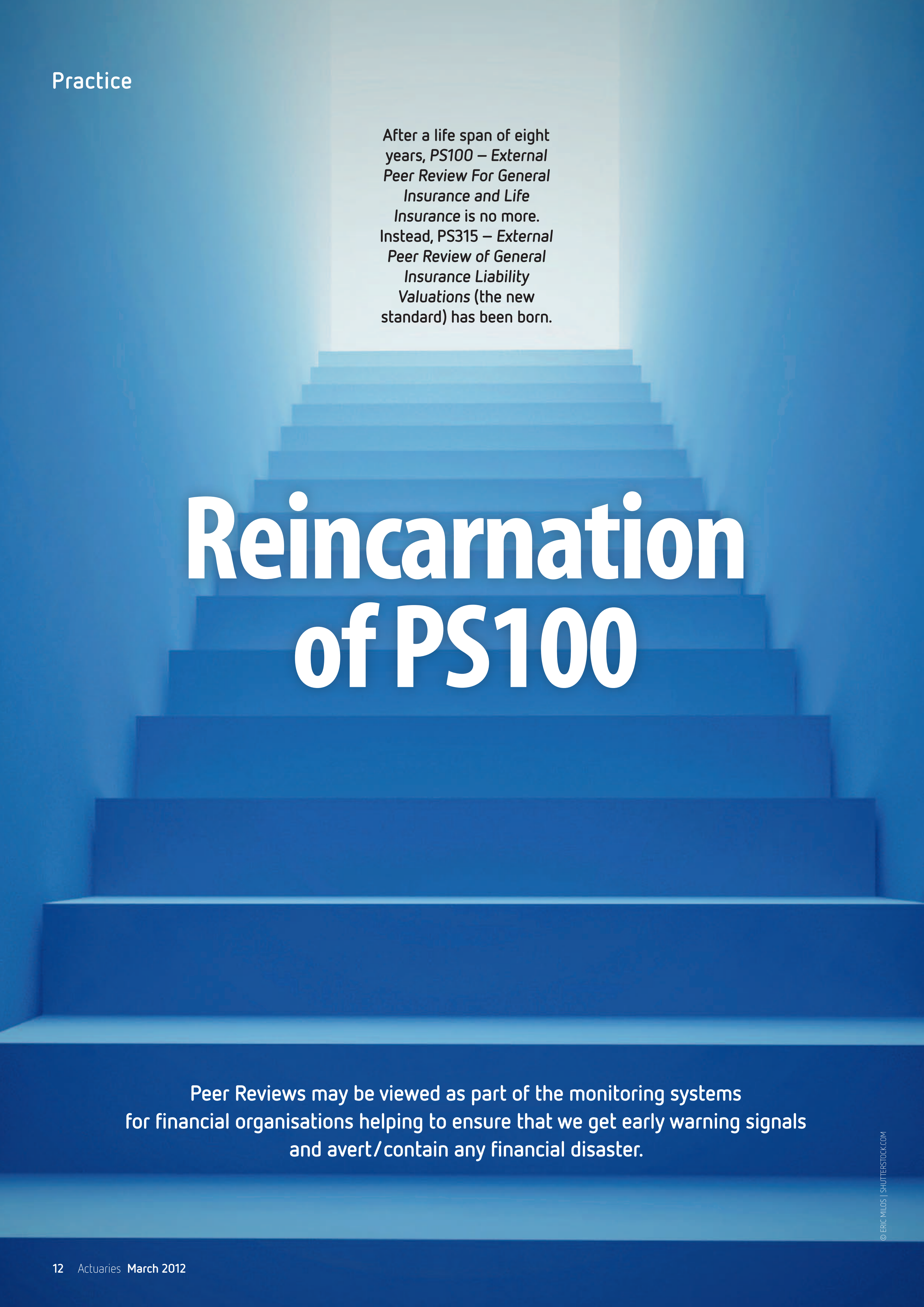
QUESTION 15: ARE THERE ANY OTHER EXAMPLES OF NERDISH BEHAVIOUR THAT YOU WOULD LIKE TO CONFESS TO?

Finally, respondents were given the chance to disclose any other nerdish things they have done. There were some unusual responses, for example • I'm a level 85 blood elf paladin trapped in the confines of a human body • Buying high tech toys for grandchildren but keeping them for myself • Having a Super Mario themed wedding • Building medieval siege weapons.

However, the majority of the answers to this question were more suggestive of the fact that the respondents were well suited to being actuaries than that they were nerds. These include • Applying the Control Cycle technique to life • In high school, I used proof by induction questions as study rewards... sometimes I still do • Loving maths/stats/Excel/studying • I took Part 3 study notes with me into an Asian night club • Using Excel to model a mortgage or plan a holiday • Memorising pi to hundreds of decimal places (the highest number claimed was 600) • Writing an actuarial paper called *Extra-Terrestrial Influences* on *Nature's Risks.*

CONCLUSION

Based on the responses received to this survey, it appears that many actuaries believe that both they and their peers are nerds, but few of these people fit into the traditional nerd stereotype of friendless, *Star Trek*-quoting geeks. Instead, what was observed was a group of intelligent people who just happen to have an unusually strong interest in maths and related subjects, an interest that makes them more likely to succeed in their chosen profession; and perhaps herein lies the secret to the recent upsurge in the popularity of nerds. The nerds' intense interest in complex subjects sets them up for success, and succeeding in a complex discipline is impressive. No wonder so many people now want to be nerds.



After a life span of eight years, *PS100 – External Peer Review For General Insurance and Life Insurance* is no more. Instead, PS315 – *External Peer Review of General Insurance Liability Valuations* (the new standard) has been born.

Reincarnation of PS100

Peer Reviews may be viewed as part of the monitoring systems for financial organisations helping to ensure that we get early warning signals and avert/contain any financial disaster.

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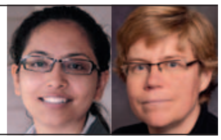


One of the key changes is that Life Insurance is no longer under the scope of the new standard. However, peer review (internal or external) is relevant in both life and general insurance areas because:

- Important financial decisions can be based on actuarial work, which can often be complex. A peer review can help improve the quality and consistency of the actuarial work and hence the decisions.
- A peer review guards against "sub-conscious optimism" – be it in pricing or reserving – that, much as we impartial professionals might like to deny it, we are all at risk of.
- Peer reviews can help actuaries leverage off each other's experience not only to increase knowledge and skill levels but also to gain support in the event of complex/contentious issues or commercial pressure.

In this article, Bridget looks at the application of Internal Peer Review in Life Insurance through an implementation example and Suruchi outlines some useful practices in the context of External Peer Review (EPR) in the General Insurance sector vis a vis the new standard. The discussion points in this article are not new. However, by broaching these topics we hope to provide some insights on how to make peer review a valuable process.

HOW DO YOU CURRENTLY USE PEER REVIEW IN YOUR LINE OF WORK?

If you treat peer review as a box-ticking, butt-covering compliance exercise, it is almost certainly the waste of time and money that you expect it to be. However, I have been agreeably impressed by the value that you can get from a peer review process implemented in a meaningful way.

The reorientation of the Professional Standard to EPR for General Insurance only shouldn't be taken to mean peer review is any less important in Life Insurance.

In my case, I inherited a situation where every piece of work on one site was peer reviewed and on the other site nothing was. Both sites were doing exactly the same work: life reinsurance pricing exercises in Continental Europe. Clearly neither site was operating ideally. Something needed to be done: I was responsible for the quality of the pricing, but the pricing actuaries didn't work directly for me. I looked around at international practice for inspiration and found that the Australian PS100, as it was at the time, was by far the most constructive reference. In other jurisdictions, peer review seemed either very compliance oriented and excessively formalised, or indeed still at discussion stage. Nevertheless, starting with PS100, there were still a few enhancements required in order to make the process fully effective, at least in an internal context.

Peer review is ultimately a cost/benefit trade-off – it's only worth investing the effort if the likely "savings" generated is greater – be those savings reputational or financial. In standardised or small deals, this is unlikely to be the case. So, in an internal peer review process, triggers become important. What sort of exercises warrant a peer review and which don't? This can be difficult to quantify, but the management should be comfortable with what does/doesn't get a "second opinion". The types of triggers we used included the obvious (e.g. financial exposure) and the less so (e.g. the novelty of the risk and/or market and our depth of experience or level of comfort).

Another area where PS100 was silent is the resolution process when the Reviewing Actuary cannot give a "clean" opinion. Firstly, we needed to understand the sensitivity and maturity required to write a critical Peer Review report. It shouldn't be in any way combative or attacking. Indeed, the Reviewing Actuary needs to have great credibility and demonstrated independence so that others can be confident that there is no bias or risk of personal attack. Obviously, the first step was to see the producers of the work concerned, to discuss and ensure no misunderstanding: giving them the chance to correct if needed. However, if it comes down to a difference of opinion, we found that we needed to define a simple referral process for resolution, to make the process whole and workable on an ongoing basis.

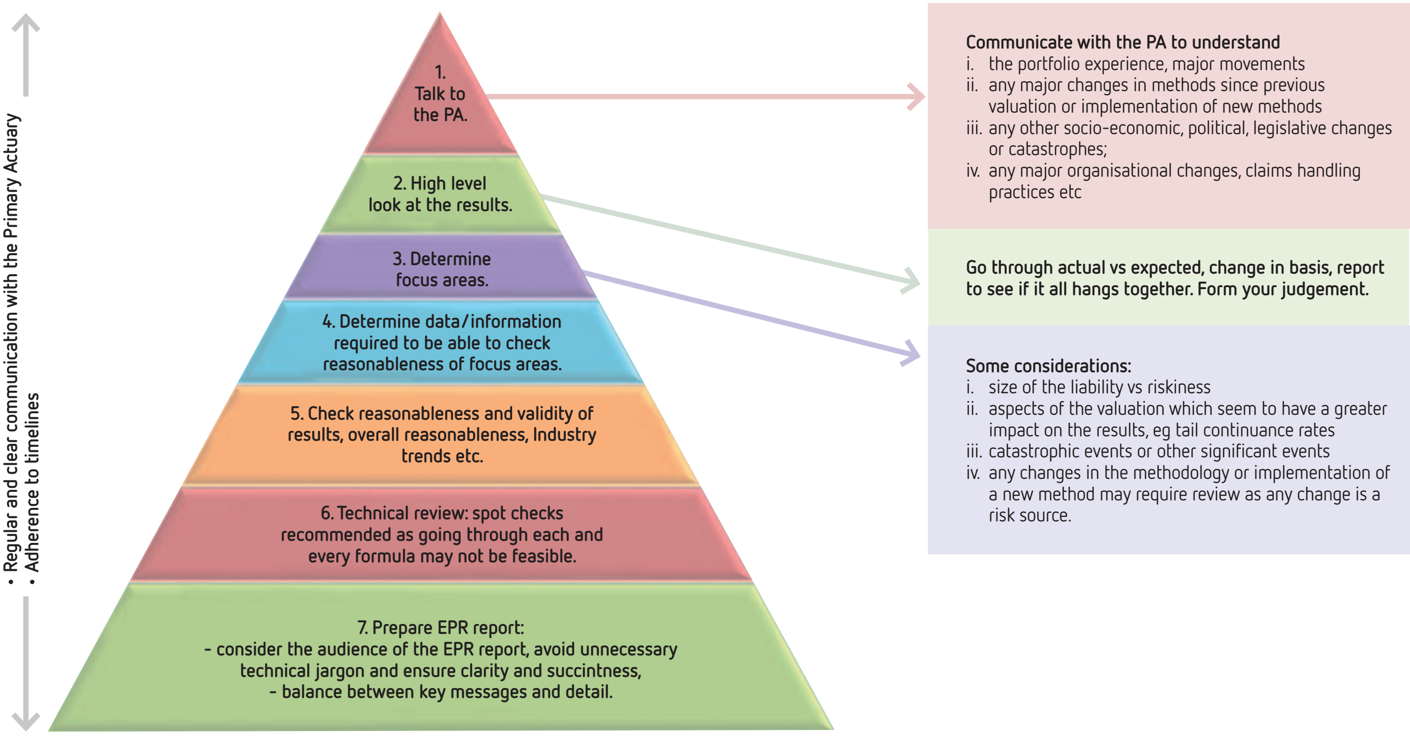
Surprisingly, though, I also saw teething problems with the fear of giving a clean peer review. I saw this happen with an experienced person who was new to a particular area of practice. This led to nit-picking and irritation all round. Since it is not at all constructive, it needs to be dealt with promptly whenever it arises.

We found that training was a vital component of developing acceptance and the effective use of Peer Review as we envisioned it. This was made "real" via the use of case studies – including writing Executive Summaries. Defining clear roles and responsibilities also helped. One particularly key understanding was the importance of clearly separating doing, checking and reviewing. Checking and peer review are not the same – not just because they should be performed by different sorts of resources, but because they provide different sorts of assurance: the first is more focussed on accuracy, while the second is centred on reasonableness and overall coherence.

We also decided to alternate the roles of Primary Actuary (PA) and Reviewing Actuary (RA) as often as feasible. This has a number of benefits: it is a way to train and transfer knowledge; it reduces single person reliance issues; and more importantly, it reduces the "ivory tower" risk. Of course, care is always needed to maintain independence.

Examples of benefits we experienced:
Simple benefit: Pricing actuary priced without brokerage. Marketing actuary thought price included brokerage. Too small for the formal detailed checking, so only Peer Review (due to size of exposure) picked it up. Saved EUR 100,000.
More subtle benefit: maturing experience for early career staff in how to deliver a differing opinion. This is not really developed in the Continental European actuary's toolkit.

Time pressure is always with us, and one view says Peer Review just adds unnecessary delay and length to an already stressed timetable. However, by creatively thinking about how to move from series to parallel processing for certain components, issues can be identified and dealt with early in the process rather than losing time redoing all steps. The whole task can be executed more quickly while maintaining high quality.



Lastly, when a peer review makes recommendations, these may be specific to the particular exercise or more generally applicable. It's worthwhile finding a way to capture these wider recommendations and implement in due course, thus not losing the benefit they may bring.

I hope this example of implementing a peer review process will be of use to those facing similar issues. I welcome any discussion on best practice in this area.

HERE'S LOOKING AT YOU, PS315.

The above Life Insurance example talks about points that are important to make a peer review process, be it internal or external, a beneficial one. This section highlights these in the context of EPRs in General Insurance with excerpts from the new standard where relevant.

Based on my conversations with various actuaries, I have summarised the basic steps of an EPR in the diagram below. This can, at most, be taken as high level guidance, not a complete manual for the EPR process. Key points are as follows:

- Open, clear and regular communication is an important aspect of an EPR process. Talking to the PA is important, not only to gain an initial understanding of the portfolio, but also to keep the PA informed of any thoughts and opinions as and when they are formed. This helps to avoid misunderstandings and resolve issues sooner rather than later.

> **Clause 5.3.2 of PS315 –** ...there can be differences in professional opinion between the Primary Actuary and the Reviewing Actuary. In such situations, there is an inherent need for communication between the Reviewing Actuary and the Primary Actuary. Recognising this, at the earliest opportunity, the Reviewing Actuary must discuss any such issues arising with the Primary Actuary.

- An EPR need not be a detailed "tick-a-box" checking process. More often than not it is ensuring that the work is reasonable. To be able to do this, determining the focus areas is important. Step 3 of the diagram shows some aspects that can be considered.

> **Clause 1.4.1 of PS315** The purpose of External Peer Review is to review and provide a conclusion on the reasonableness of the Primary Actuary's actuarial advice.
> **Clause 7 of PS315** lists the matters for consideration by Reviewing Actuary.

- The level of detail and queries in an EPR process may be important in the interest of time and cost. To some extent, the detail and queries may be proportional to the experience of the RA. While a veteran actuary may know exactly what to look for, a suitable but less experienced actuary may have to sift through more detail before being comfortable with the work. The RA needs to remember that the aim is not to re-do the work, and that while no question is a bad question, it could be an immaterial one.

Whether mandatory or optional, the only way to get value from the process is to expect to get it and build it out accordingly.

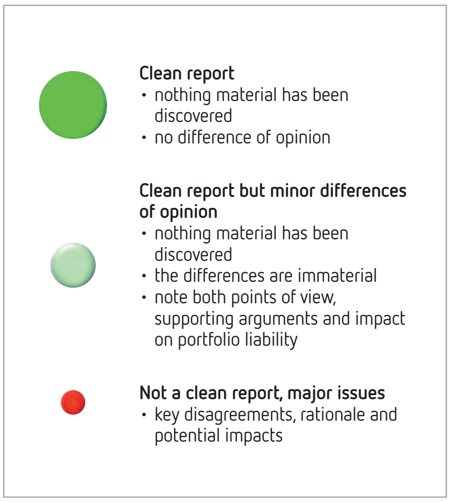
Clause 4.1.1 of PS100 - Members must satisfy themselves that they have the relevant expertise and experience consistent with the Institute's Code of Professional Conduct. (This is not explicitly stated in PS315 as it seems unnecessary duplication of Code requirement).
Clause 5.3.2 of PS315 - In doing so, the Reviewing Actuary must be aware that he or she is reviewing the work that has been undertaken, rather than performing a Specified Valuation.

- For an EPR to be effective it needs to be given adequate priority and time (this requires strong commitment from management). In cases where it is anticipated that there is not enough time, a parallel review methodology (as discussed in the life insurance example) can be followed.

- When deciding the length and content of an EPR report, it is important to keep the following in mind:
  - The audience of the report – Board/ management/ PA;
  - Clarity and succinctness of the message; and
  - Criticism, if any, is constructive and not demeaning.

Clause 8.2.2 of PS315 - The level of detail to be provided in actuarial reporting will depend on the size and complexity of the Entity, and considerations of Materiality.
Clause 8.2.3 of PS315 - It is acceptable for the Member to prepare a short form report (for an audience such as the Board, as described under PG 199.01).

A suitable approach may be a short report along with an appendix.The former containing the key messages while the latter giving details of all that was reviewed, additional improvement ideas etc. The following diagram broadly shows the possible outcomes of an EPR and gives examples of key messages for each outcome.



We expect that the vast majority of outcomes would, in practice, be the first two. However, in all cases, communication with the PA with regards to the findings should already have taken place and issues resolved to the extent possible, with the aim of avoiding being in the red zone.

- The form of assurance (negative or positive) required should be decided at the outset of the EPR. The scope of most EPR's will generally require a negative form of assurance as a conclusion.

Clause 8.3.1 of PS315 gives the following specimen wording (a form of negative assurance):
"Having carried out the review as described in this report, nothing has come to my attention that would lead me to believe that the Primary Actuary's Specified Valuation results are unreasonable."

However, there may be situations where the stronger form of positive assurance may be required. Ultimately, it will depend on the purpose of the peer review. There is generally an implicit trade-off between the additional cost of providing a positive assurance conclusion and the purpose of the EPR.

Understanding Positive vs Negative Assurance:
Positive assurance – an affirmative statement/opinion given by the auditor (eg. the financial report taken as a whole is free from material misstatement, whether due to fraud or error).
Negative assurance – a statement indicating that, on the basis of the procedures performed, nothing came to the auditor's attention that causes the auditor to believe that the financial report does not present fairly, or if applicable, is not true and fair, in all material respects, in accordance with the applicable financial reporting framework.

FINALLY

In New Zealand there are extensive systems to monitor seismic waves to anticipate earthquakes so that timely action can be taken to protect people. Peer Reviews may be viewed as part of the monitoring systems for financial organisations helping to ensure that we get early warning signals and avert/contain any financial disaster. On a less dramatic note, whether mandatory or optional, the only way to get value from the process is to expect to get it and build it in accordingly. So, make the most of it. A

Further readings

1. American Academy of Actuaries – Council on Professionalism:
Peer Review Concepts on Improving Professionalism, Discussion Paper Prepared by Committee on Professional Responsibility
2. Peer Review Paradox by Lawrence Tsui and Jia Na

Chris Ryan John Edwards Tony Davis Alex Twigg Brian Bissaker

THE ACTUARIES INSTITUTE WILL CONVENE THE 2012 FINANCIAL SERVICES FORUM IN MELBOURNE FROM 30 APRIL – 1 MAY 2012

The theme for the Forum is Think Big. We must "think big" if we are to successfully navigate the short-term turmoil whilst preparing for the long-term structural changes facing Australia and the global community. How can we manage the risks, capitalise on the opportunities and push the boundaries to continue to add value to our industry and to the millions of Australians who entrust their money to us?

FSF2012 will have a diverse program featuring six plenary sessions and 48 concurrent sessions, covering a wide range of topics including Structural Reform & Public Policy, Driving Value in Financial Services, Superannuation & Wealth Management, Enterprise Risk Management & Corporate Governance and Connecting with Consumers. The Forum attracts attendees from the banking, investment, finance, financial planning, life insurance, superannuation, wealth management and risk management industries.

This is a timely and important opportunity for you to join your actuarial colleagues and other professionals to network and keep abreast of the latest developments in the financial services industry in Australia and internationally.

Register Online www.actuaries.asn.au/FSF2012

PLENARY SESSION SPEAKERS

KEYNOTE: THE BIG PICTURE

Take part in a broad discussion of the industry we inhabit, the global forces for change and likely reactions by market participants.

Todd Sampson, CEO, Leo Burnett Sydney
Todd is the CEO of Leo Burnett, Australia's top creative advertising agency. In 2012, they were ranked as the 7th most creative agency in the world and the only Australian company to ever make the top 10. Todd is also the co-creator of the Earth Hour initiative which has been recognised as one of the best ideas globally with over 1.4 billion people taking part in over 5000 cities. The Financial Review ranked him as one of the most influential leaders in the country. Todd is also the breakout star of the hit ABC television show The Gruen Transfer and a co-host of Channel Ten's The Project.

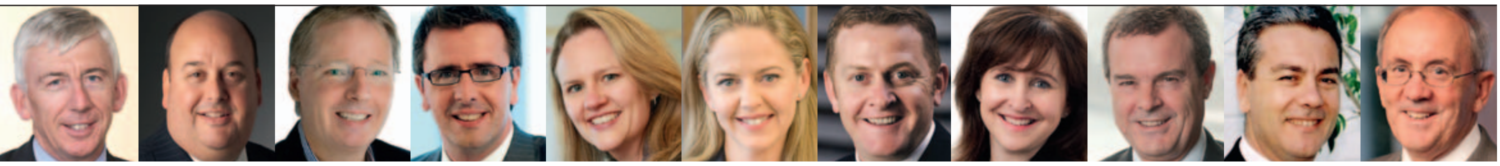
PLENARY ONE:

CHARTING A COURSE THROUGH THE GLOBAL ECONOMY

Join us as we embark on a journey through the global economy and explore the recent ructions and ramifications for the Australian financial services industry

Dr John Edwards, Visiting Fellow, Lowy Institute, and Member of the Reserve Bank Board
John Edwards is a Visiting Fellow at the Lowy Institute, an Adjunct Professor with the John Curtin Institute of Public Policy at Curtin University, and a member of the Board of the Reserve Bank of Australia. From 2009 to 2011 he was Director for Economic Planning and Development for the Economic Development Board of the Kingdom of Bahrain. From 1997 to 2009 Dr Edwards was Chief Economist for Australia and New Zealand for the global financial group, HSBC. In 2008 he was given leave of absence from HSBC to accept a secondment to the Australian Treasury as Chief Adviser, Financial Markets.

Chris Ryan, Former head of Asian and Australian Investment Management Companies
Chris Ryan has worked in financial services for more than 30 years, with almost half of his career outside Australia in senior management positions in Asia. Most recently, he was CEO and Managing Director for Perpetual Limited in Sydney, Prior to that, he was Head of Asia for Fidelity International Limited, based in Hong Kong and before that he was CEO of ING Investment Management Asia Pacific, also based in Hong Kong. He has extensive experience across all Asian markets, including China, Japan, Taiwan, Korea, Hong Kong, Thailand, Malaysia, Singapore, India, Philippines, Indonesia, Pakistan and Dubai. He established the first Chinese joint venture fund management company in 2001 and in addition to his investment management company experience, has also been a director of a number of Asian life insurance companies.



Tony Lally Marc Lieberman Jeremy Cooper Michael Pennisi Pauline Blight-Johnston Melanie Evans Paul Barrett Debby Blakey Andrew Gale Greg d'Arville Dick Morath

PLENARY TWO:

THE DIGITAL REVOLUTION

The adolescent digital world of the 90s has grown up and matured to a savvy force for change. This plenary will provide you with access to some of the latest thinking in the digital space and examples of how financial services firms are employing this medium to transform the way their businesses operate.

Tony Davis, Director, Quantium
Tony Davis leads many of Quantium's marketing, media and customer analytics programs and partnerships. With a background in marketing and strategy at financial services firms including NatWest in the UK, Aussie Home Loans, and a wide range of FS clients since joining Quantium 10 years ago, Tony has strong experience in the sector. Quantium has a very clear picture of how the digital world is affecting the Australian landscape. This includes the impact of digital strategies in customer acquisition, tracking the competitive environment and driving new levels of customer engagement.

Alex Twigg, General Manager, UBank
After 12 years in retail banking and mortgage lending in the UK Alex joined the Egg Bank launch team in 2000. In 2004, Santa Fe, Egg's next generation banking platform subsidiary was acquired by Keane Inc and Alex lead his team to New York to create Keane's Customer Experience Technology Practice, helping to shape the direct banking and delivery strategies of Fidelity, Citigroup, Etrade plus many other Fortune 500 clients.

PLENARY THREE:

REACTIONS TO RETIREMENT

Combining some of the pre-eminent industry experts in the retirement and superannuation industry, we will discuss the impact of the approaching demographic tidal wave and it's implications for member engagement, product development and public policy.

Brian Bissaker, CEO, Colonial First State
Brian joined Colonial First State in 2002 and was promoted to Chief Executive Officer in 2006. Prior to joining Colonial First State, Brian spent eight years at BT Funds Management where he was an Executive Vice President and Head of Product. Brian commenced his career as a taxation and superannuation consultant at KPMG servicing the finance, funds management and superannuation industries.

Tony Lally, CEO, Sunsuper
Tony has been CEO of Sunsuper since 2007. He has overseen a number of strategic initiatives including the acquisition of CitiStreet Australia which resulted in the in-sourcing of all administration and member services for the fund. Tony has over 30 years senior executive experience in the insurance, superannuation and wealth management industries in a number of countries. He spent five years based in Japan with Deutsche Bank and Deloitte where he developed and implemented strategic plans for funds management, actuarial and insurance businesses throughout the Asia-Pacific region. He was also with Commonwealth Bank for 10 years where he led Australia's largest retail funds management business.

Marc Lieberman, CEO, MetLife Australia
As CEO, Marc is responsible for all aspects of MetLife's Australian business with a strong focus on driving profitable long term growth. Marc joined MetLife in April 2010 and under his leadership, the focus will be to drive a customer service and value driven organisation. With 25 years experience in financial services, Marc has extensive experience in delivering high growth results through his strong leadership in his capacity as CEO.

Jeremy Cooper, Chairman, Retirement Income, Challenger
Jeremy Cooper is Chairman, Retirement Income at Challenger. He is a lawyer by training with experience across a broad range of financial services disciplines. Prior to joining Challenger, Jeremy was appointed by the Federal Government to chair a wide-ranging review of Australia's superannuation system, now known as the 'Cooper Review'. Before the role, Jeremy was deputy chairman of ASIC for five years from mid-2004. He is also a member of the Finsia Policy Advisory Council and the Industry Advisory Committee of the Australian Centre for Financial Studies.

PLENARY FOUR:

GIMME GIMME... A GUIDE TO CONSUMER NEEDS

Customer needs are core to the financial services industry. Join the panel as they discuss what consumers really need how we might manage to deliver it.

Michael Pennisi, Chief Officer Strategy, QSuper
Michael Pennisi has over 22 years experience in the Australian financial services sector having commenced his career with one of Australia's largest financial services firms. During his career, Michael has held senior executive positions with some of Australia's largest superannuation funds, fund managers and financial planning firms and his experience broaches the private and public sector, including a role with the Insurance and Superannuation Commission. He has been an active contributor to industry associations and has held State executive positions for ASFA and is a former member of the FSC Global Markets Board Committee.

Pauline Blight-Johnston, Managing Director, RGA Reinsurance Company of Australia
Pauline Blight-Johnston, Managing Director of RGA Reinsurance Company of Australia Limited is responsible for RGA's business in Australia and New Zealand. Pauline has over 18 years of actuarial and financial management experience in the life insurance industry, both in Australia and internationally. Prior to joining RGA in 2007, she was Chief Financial Officer and Appointed Actuary for Asteron Life. Before then, she was a consulting actuary with Tillinghast and an investment banker with Morgan Stanley Dean Witter.

Melanie Evans, Head of Super and Platforms, BT Financial Group
Melanie Evans has recently been appointed to the role of Head of Superannuation and Platforms at BT Financial Group. She is responsible for a business portfolio spanning Corporate Super, Retail Super & Investments, BT Super for Life as well as the Asgard & BT Wrap Platforms. Prior to this appointment, Melanie was BT's Head of Superannuation. She has also headed up BT's Personal Super business, where she successfully spearheaded the launch and distribution of their innovative and award-winning online superannuation product – BT Super for Life. At BT, she has also held roles including Head of Brand and Head of Group Marketing.

"Given the different plenary speakers it is an opportunity to hear commentary from high profile leaders on the current environment and the potential commercial and other implications for the industry over the next few years."
– previous delegate

Register Online www.actuaries.asn.au/FSF2012

PLENARY FIVE:

THE HARD SELL

The advice industry is under enormous scrutiny and strain. This plenary will explore the changes taking place and discuss the evolution of advice within the Australian financial services industry.

Paul Barrett, General Manager Advice & Distribution, ANZ
Paul Barrett has over 17 years experience working in banking and wealth management and is currently the General Manager of Advice & Distribution for ANZ Wealth. Prior to this Paul was the General Manager of Colonial First State Advice Business. In this role he was responsible for the Commonwealth Financial Planning, BW Financial Advice, Financial Wisdom, Whittaker Macnaught and Advice Essentials dealer groups, and the related dealer support areas. Before this he was the General Manager of Colonial First State Distribution which included responsibility for distribution, corporate superannuation, CFS products and services to the IFA market, the Financial Wisdom and Whittaker Macnaught dealer groups and dealerships services.

Debby Blakey, Deputy CEO, HESTA Super Fund
Debby was appointed to the role of Deputy CEO at HESTA in January 2012, prior to which she was Executive Manager – Member Advice from September 2008. Debby brings more than 20 years experience in superannuation and financial planning to HESTA. She started her career working on defined benefit mergers and valuations for a consulting actuarial practice in South Africa, before setting up her own business as an employee benefit consultant with a particular focus on superannuation. Debby has previous retail and industry fund experience in business development, operational management and member advice. She is passionate about delivering quality advice to members who have historically not had access to cost-effective advice.

Andrew Gale, Former CEO & Managing Director, Count Financial
Andrew Gale was Managing Director / Chief Executive Officer of Count Financial (Count) up until December 2011. Count is Australia's largest franchised network of financial planning/wealth accountants and advisers with around $14 billion of clients' funds and loans under advice. Count was acquired by CBA in December 2011. Andrew has over 30 years' experience in the financial services industry, including leadership of Deloitte Actuaries & Consultants and senior executive roles with two of Australia's leading financial services organisations – MLC and AMP. Andrew is currently enjoying a short sabbatical.

PLENARY SIX:

LEVERAGING ENTERPRISE RISK MANAGEMENT TO SHAPE STRATEGY – A BOARD PERSPECTIVE

This session aims to demonstrate how valuable it is for a Board to explicitly link risk management and strategy. It provides the board members' perspective (i.e. top-down) to this modern day enterprise challenge.

Greg d'Arville, Principal, crgESSENTIALS
Greg d'Arville is an educator, journalist and marketer with over twenty-five years of compliance, governance and risk management experience. Greg delivers training, audit and consultancy services to board members, executives and their direct reports through his Melbourne-based company crgESSENTIALS.
He has advised numerous companies and government business enterprises on governance, strategy and risk issues.

Dick Morath, Chairman, GWM Advisor Services
Dick's current roles include; Chairman of the Advice & Licences Boards and all Financial Planning companies in NAB / MLC in Australia, and of the Compliance Committee for all NAB / MLC dealerships; Director of MLC Nominees, the trustee for all the superannuation business of MLC; Chairman of Plum, the MLC group vehicle providing member services to the large corporate superannuation market. Director PFS Nominees, the Plum Trustee; Chairman of National Australia Trustees, part of the National's Private Bank; Chair of NAB Health Advisory Council, the advisory board to the NAB Health business, which is part of the Business Bank; Director of JANA Investment Advisors Limited; Director of BNZ Life and Director of ASX listed Platinum Capital Limited and Chair of that company's audit committee.

NETWORKING

There will be several opportunities to network at FSF2012 throughout each day and at the Cocktail Party on Monday evening 30 April. This function presents a perfect opportunity to relax, catch up with colleagues and meet new people.

CONCURRENT SESSIONS

The presenters of the concurrent sessions have all risen to the challenge of Thinking Big with topics ranging from reforming the age pension, through measuring house prices, managing operational risk to thinking big by thinking micro. The topics are very representative of each strand of financial services covering financial planning, life insurance, health insurance, banking, superannuation and retirement. Risk management and investments are also well covered. The concurrent sessions will provide financial services professionals with a range of thought provoking options to choose from.

For those interested in superannuation, several presenters will give their thoughts on how to fund retirement. Other presenters have focused on APRA's reforms, risk management developments in super and how to share investment performance among members.

Those with a focus on life insurance can discuss with their peers disability income experience, the LAGIC (Life and General Insurance Capital) reforms to capital, pricing practices and new analytical approaches. Those with an interest in health insurance can consider capital for mutual health funds and how to fund health costs in old age. The banking community can debate the challenges of cross selling, home pricing and related default risk and how to manage liquidity.

Risk managers and investment experts also have plenty to choose from. There are several case studies including coverage of governance for not for profits and the history of The Equitable. Other papers consider the role of the actuary in enterprise risk management and operational risk. The important topic of risk culture is also considered and the role this plays in risk management. Several of the investment topics consider risk management in practice for hedging mortality and investment risks and for driving profit when all returns are low.

It will be challenging for delegates to choose between such an extensive range of topics. Fortunately, the Financial Services website will enable you to catch up on those sessions that you have missed and there will be plenty of time during the forum to quiz other attendees on the sessions that you missed (if not the presenters themselves!).

Register Online www.actuaries.asn.au/FSF2012

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Topic Snapshot

Thinking Big

Living until 120: the implications of absolutely everything
Fundamental Change in the Global Economy and Its Impact on Australian Actuaries
How to Think Big
Fearless Leadership – where Neuroscience meets Leadership
Think Big by Thinking Micro
Key Findings of a Recent Survey of Pricing Practices

Life Insurance

Disability Income – "Have We Learnt Anything?" and "Where to Next?"
ICAAP – "Can This Spawn Innovation?" and "What Should Actuaries Do About It?"
Group Insurance - Where Does the Operational Risk Lie?
Implementing Daily Solvency Monitoring – a Case Study
International Solvency Regimes for Annuity Markets

Superannuation and Retirement

Five Thoughts on Australia's Ageing Population
Reforming the Age Pension
The Lifetime Harvesting Plan: Annuities with Sharing of Mortality and Investment Risks
APRA Prudential Standards for Superannuation
Crediting Rates or Unit Prices – Lessons Learned in These Volatile Times
Actuarial Challenges in Defined Contribution Schemes
Are Defined Benefit Funds still Beneficial?

Anayltics and Marketing

Financial Services Applications for Customer and Business Analytics
Google Insights for Financial Services
Predictive Analytics Applications in Retail Banking
Cross-Sell: The Growing Need to Diversify
Frequent flyer analytics – The link between loyalty and financial services

Financial Planning

Financial Planning Frameworks of the Future
Implications of the FOFA Reforms
FOFA Opportunities and Challenges

Investment and Markets

Financial Services Market Update – Post GFC
Latest Developments in Risk Management, Hedging and Capital Markets for Insurers
How Capital Markets Can Help You Manage Meta-Risks?
Squeezing the Lemon – Driving Life Insurance Profitability in a Low Return Environment
The Role of the Infrastructure Asset Class in Superannuation Funds in 2022
The Lifecycle Investing Debate
Catastrophic Mortality Bonds: An Effective Hedge?

Risk Management

Developing Risk Management Governance at a Not for Profit – a Case Study
The End of the Equitable – a Case Study
ERM: Theory, Practice and the Actuary
Developments in the Application of Complex Systems to ERM and Actuarial Work
Measuring and Managing Risk Culture
Operational Risk + Actuaries = Big Opportunity + Big Challenge

Banking

Fresh Insights – Australian/NZ Home Lending Default Risk
House Price Risk Models
Liquidity Risk Management

Health Insurance

Capital Issues for a Mutual Health Fund in Australia
Health Costs, Ageing, Superannuation

Register Online www.actuaries.asn.au/FSF2012

Genevieve Hayes
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"I have discovered a truly marvellous proof of this, which this margin is too narrow to contain" – Fermat.

Around the world in ten puzzles

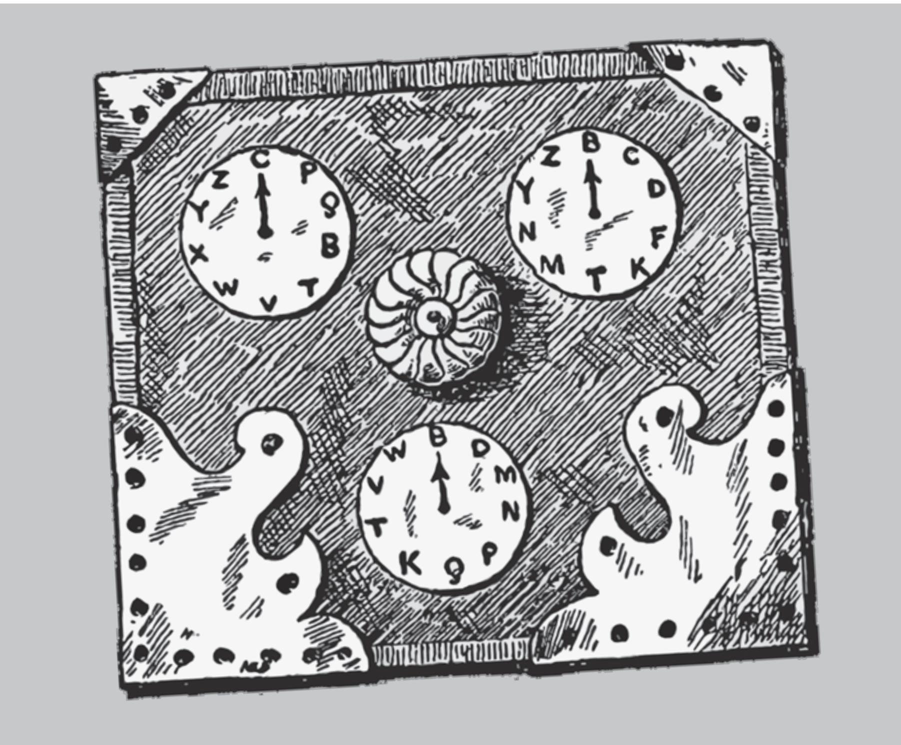
Hello and welcome back for 2012. This year In the Margin is going to take readers Around the World in Ten Puzzles. Each month we will visit a different location, where we will find a different puzzle. These puzzles are in no way connected beyond the central theme of "Around the World".

This month we are going to begin our puzzle tour by visiting San Francisco, USA.

ESCAPE FROM ALCATRAZ

Located on a rocky island in San Francisco Bay, Alcatraz was America's first maximum security civilian prison and operated as such from 1934 to 1963. During its 29 years of operations, it housed some of America's most notorious criminals, including Al Capone. Only 36 prisoners ever attempted to escape and of those men, 23 were caught, six were shot and killed and three drowned.

Before becoming a civilian prison, Alcatraz was a military prison, housing Civil War prisoners from as early as 1861. What many people do not know, however, is that back in the 1860's, prior to the first prisoner arriving, much work went into determining the best locks to use on the cells. One particular lock that was considered comprised three dials with ten letters on each dial (pictured below). Moving clockwise from the top left, the letters on the three dials are – Dial 1: C, P, Q, B, T, V, W, X, Y, Z; Dial 2: B, C, D, F, K, T, M, N, Y, Z; Dial 3: B, D, M, N, P, Q, K, T, V, W.



To stop the prisoners from cracking the lock by trying each of the 1000 combinations, this particular lock was fitted with an alarm that would sound on the third inaccurate attempt. Nevertheless, the lock was ultimately rejected as being too simple to crack. Given the fact that the three letters in the correct combination spell a word in English, can you deduce the secret word?

For your chance to win a $50 book voucher email your solution (with reasoning) to: inthemargin@actuaries.asn.au

NATIVE MATH (AA165 SOLUTION)

The twelve formulae given in the puzzle are:

1. The formula for the EPV of a whole of life annuity;
2. exp(iπ) = -1;
3. The pdf of the normal distribution;
4. The area of a circle;
5. The put-call parity formula;
6. $\Gamma(x) = (x-1)\Gamma(x-1)$;
7. The quadratic formula;
8. CAPM;
9. The Black-Scholes differential equation;
10. The correlation formula;
11. The formula for a straight line; and
12. The eigenvalue equation.

By identifying these equations, the letter mapping between our language and the "native language" can be derived as follows: A = V, B = S, C = P, D = X, E = λ, F = M, G = ∞, H = F, J = Γ, K = A, M = σ, N = β, O = π, P = μ, Q = R, R = ∂, S = C, T = B, U = Y, V = I, W = E, X = ρ, Y = T.

WORDOKU (AA166 SOLUTION)

As is suggested by the title to the puzzle, the first puzzle given in AA166 is a Sudoku puzzle with letters instead of numbers. The solution is:

T	O	Y	B	W	E	A	R	D
W	D	E	O	A	R	T	Y	B
B	R	A	D	T	Y	W	O	E
A	B	R	Y	E	W	O	D	T
E	Y	O	T	R	D	B	W	A
D	W	T	A	B	O	R	E	Y
Y	E	B	W	O	T	D	A	R
R	T	W	E	D	A	Y	B	O
O	A	D	R	Y	B	E	T	W

Six correct answers were submitted to Native Math. The winner of this month's prize, selected randomly from among the correct entries, was Jonathan Shen, who will receive a $50 book voucher. The solution to the 2011 metapuzzle will be given in next month's magazine.

Gae Robinson
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Gae answers serious and not-so-serious questions about life in the office, career, study and coping as an actuary in the real world

Ageing gracelessly?

1. 'How old are you before you stop being a 'Young Actuary' and become ineligible to do 'Young Actuary' things? Should there be something in between Young Actuary and the RAGS programs to make the 'Mature Actuaries' feel included?

2. At a recent actuarial gathering, I was referred to as a 'younger actuary' due to my youthful looks. I'm no spring chicken, so I'm not sure if that was a compliment or an insult. Can you suggest techniques to look my age, and gain respect and credibility?

We're all so troubled about age and ageing these days, aren't we? And whereas many difficult issues in life can be dealt with and put to the side, we can't do this with our age because it keeps changing! I know I can't settle on how I feel – one day it's "oh, what do I care, it's just a number." and the next I'm plunged into despair by looking in the mirror. Actually, even worse than that is how old my *friends* are looking.

And there are so many options for those of us who *don't* want to look our age – we can dye our hair, have surgery, succumb to botox… all of which represents another formidable barrier to us just dealing with it. (Should I, shouldn't I? Why is she still looking so youthful – is it for real? And so on.)

While we're talking about age, I'll also say that I am well and truly fed up with being told: "Oh, I know all about the way you think and feel because you're XX Generation". Ageist and insulting, pure and simple! If I want to judge my fellow human beings using overgeneralised pseudo-scientific classification criteria, I'll turn to astrology, thanks.

You may be thinking this all sounds like the desperate ravings of a woman whose <blank>tieth birthday is only a few weeks away…well, congratulations Sherlock!

So, to the question… how old can a Young Actuary be? The Institute's website says:

"The Young Actuaries' Program is an initiative offering seminars, workshops and discussion forums to assist *young members* with their careers. If you are *still studying or qualified in the last few years*, you will benefit from the program of events."

Carefully worded – with no mention of age of course. Are they going to refuse entry if an oldie appears? I suppose if YAP events start getting hard to get into, it's possible you'll start needing to show ID at the door. I note, though, that if you keep failing your exams, the Institute will still consider you Young even at an advanced age!

In his play *As You Like It*, Shakespeare maps out the 'seven stages of man', which can be summarised as:
Infant ("mewling and puking"), **Child** (school), **Lover** (sighs, singing), **Soldier** (builds reputation, gains recognition), **Justice** (enjoys wisdom, prosperity, status), **Old Man** (decline), **Very Old** (dementia and death).

Now, how about the seven stages of an actuarial career? They might look something like this:

Beginning (YAP) – studies Part III's, bravely updates spreadsheets without understanding their intricacies, quality of life at mercy of colleagues.
Emerging – completes exams, designs spreadsheets which mystify new graduates, manages projects.
Consolidating – finally feels they understand actuarial methods, marks exams, appears at actuarial seminars (drinks in bar until 2am each night).
Established – develops new actuarial methods, presents papers at seminars (drinks till 11pm), manages people.
Mature – chairs seminar sessions (doesn't appear in bar), sets exams, may join Council if asked nicely.
Jaded – knows answers to technical questions without needing to do the modelling, knows everyone at seminars, understands there will never be a universally accepted method for measuring superimposed inflation, works until superannuation accumulation reaches adequate level.
Wise (RAGS) – forgives younger people their faults. Perspective allows thoughtful reflections on life and actuarial issues.

Between YAP at one end and RAGS at the other, there's a wide range of groups to satisfy, and the Institute's smorgasbord of activities is part of the answer. I don't think every group need a specially designed forum – but if you're feeling left out, why not gather a few likeminded buddies and see what you can come up with? JAGS, CADS, MADS, WAGS…?

And, finally, to the desire to look older. Surely this is easily solved with an unfashionable moustache, a pair of bifocals and just the right cardigan? A

Andrew Brown
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Actuaries as leaders

Rocco Mangano, left, and Danny Shuttleworth

Over the course of this year, this column hopes to showcase how actuaries are taking the lead across different industries, companies and teams. What better place to start than AMP, with the merging of AMP and AXA Australia bringing together two very significant actuarial teams. I caught up with Rocco Mangano (Group Actuary, AMP) and Danny Shuttleworth (Deputy Group Actuary, AMP) to understand their views on actuarial leadership and the initiatives they have in place to support the development of the combined team.

Rocco and Danny believe that to be truly leading, the division must demonstrate three core attributes – genuine commitment to the development and advancement of its people, proactive partnering with the business and an ongoing contribution to the broader actuarial profession. Danny observed that "the concept of the Actuarial Professional Family is really important within AMP. We are constantly looking at ways to better support and develop our people".

AMP target development of their actuarial professionals through focussing on three key attributes – communication, business acumen and professionalism. The development programs and initiatives then take into account where the person is on their career journey and what support is required to develop them to the next stage.

The Actuarial Mentoring program helps to bring this alive. Staff are provided training to become mentors, and often comment on how much they learn through mentoring others and how this helps the team to collectively understand what is happening across the business. Some mentees have been known to describe the program as one of the best things about working at AMP! The recruitment team see the program as a real positive differentiator for people considering a career with AMP.

To broaden experiences, there is a rotation program. For the graduates, there are several rotations during their first three years, while for other actuarial professionals, rotations occur around every two years. A key focus of the rotations is to promote opportunities for people to take on more challenging roles. Some staff have commented that this has brought new life to their careers.

A "New Actuaries" program also commenced in 2010, providing development opportunities to support transition to the next stage of a newly qualified actuary's career. This includes presentations from senior executives across the business, providing deeper insight to the actuaries in business strategy, distribution, marketing and customer service.

Both Rocco and Danny believe that communicating effectively with non actuaries is one of the key challenges the actuarial profession faces. Rocco commented that "Actuaries have the knowledge and training to add real value to the business in managing risks. However, we also need to be aware that insights are only insights if the business fully understands them". Through feedback from key business stakeholders, a multi year programme has been developed for the actuarial community called "Communicating with Confidence", with progressive modules and practical assignments. Participants make presentations to non-actuarial audiences as part of the action learning. Business feedback has been really positive.

HR business partners play a strong role in overseeing these initiatives. In particular, they take an active role in succession planning for the actuarial community and working with the senior actuaries to identify potential career paths for each person.

PARTNERING WITH THE BUSINESS

The role of the actuarial team within AMP is to support the business in achieving effective financial risk management, product and insurance risk management and capital management. This includes identifying opportunities where the business can appropriately take additional risks and add value.

To support this, several of Rocco's direct reports fill key Finance Business Partner roles and regularly meet with the relevant heads of business to share their insights on the marketplace, assess the experience arising on their portfolios, discuss issues that may need to be addressed and highlight new opportunities. This also helps to ensure the senior actuaries remain close to the business.

In addition, Rocco and Danny both sit on Management Committees with oversight and responsibility for the management of key business and financial risks. These include the Product and Insurance Risk Committee and the Asset and Liability Committees, chaired by the AMP Financial Services and Group Managing Directors respectively.

There are also many Actuaries within the AMP Group who are in non-traditional roles, such as Strategy, Treasury, Operations, Human Resources, Project Management, Risk Management and Legal. This allows the business to leverage the broad capabilities of the actuaries and at the same time allows the actuaries to develop a much greater understanding of the business. This enriches the whole actuarial community through these experiences.

CONTRIBUTING TO THE PROFESSION

AMP has made a long term contribution to the profession through being involved in committees, education and Council. Recent Institute presidents from AMP include Chris Lewis and Fred Rowley. Rocco and Danny have had leadership roles in key Institute committees and both believe that contribution to the profession is also an important developmental opportunity for the individuals involved as it provides them exposure to different perspectives and allows them to develop their capabilities in new areas.

AMP Actuaries who attended the Institute's Step Up Program have also been involved in a number of community based projects, such as helping high school maths students understand what actuarial maths is all about. I have made a note to attend one of those sessions!

Of course, all of this doesn't happen without the commitment of the senior team. The senior actuaries have implemented an accountability process, where each quarter they share what they have done in the last quarter to develop their people. This has built a strong shared commitment, and has also led to the sharing of ideas.

As I review this final draft (cross fingers), I feel almost exhausted reflecting on the tremendous amount of work and commitment that has been put into developing a leading culture. While Rocco and Danny refer to this as a work in progress, there is no doubt it stands the AMP Actuarial Professional Family in very good stead.

If you have an example that highlights how actuaries are taking the lead, please contact me. A

CHRISTCHURCH EARTHQUAKE, FITZGERALD AVENUE BRIDGE – © NIGEL SPIERS | DREAMSTIME.COM

NZSA General Insurance Seminar 2011

In late 2011, NZSA organised two events, the General Insurance Seminar in October and the Financial Services Forum in November. Understandably the GI Seminar was focused on the Christchurch earthquakes which have had an enormous impact in New Zealand. Apart from the tragic loss of life, the financial cost is estimated to be in excess of $NZ15bn (approximately 8% of GDP). Unlike other countries (such as Japan) the bulk of these financial costs will be covered through insurance; as a consequence this is a large challenge for the industry and the actuarial profession.

New Zealand has an Earthquake Commission (EQC) which is responsible for absorbing the 'first losses' on residential properties affected by earthquakes and other natural perils. Hugh Cowan from the EQC outlined some of the challenges the organisation is facing including:

- Dealing with an event that is five times larger than that anticipated by a recent review into the EQC;
- Project managing the repair of around 100,000 houses with a total expected cost of some $NZ3bn;
- Allocating losses between the EQC and insurers across 13 separate insurance events; and
- The unforeseen severe building damage from the impact of the earthquake on the land.

Colin Brigstock, Appointed Actuary of AMI, outlined his experiences in dealing with the aftermath of the earthquake as an Appointed Actuary. He shared with us the difficulties in estimating losses arising from the events due to the large number of uncertainties/unknowns, including:

- Understanding the IBNR pattern when insurers act as a second loss cover (to the EQC);
- The extent, quality and potential bias of the available loss estimates;
- Understanding how EQC and reinsurance arrangements play out across multiple events;
- The exposure to land remediation zones and the outcome of negotiations with the Crown;
- Demand surge – how much, when, for how long?

Dr Terry Webb and Andrew King from GNS Science gave a scientific perspective on the earthquakes. They explained that the higher than expected shaking, shallowness of the earthquake and proximity to the CBD all contributed to the damage being greater than expected given the size of the earthquake.

Ben Miliauskas from Aon Benfield discussed how reinsurers were dealing with the large number of catastrophes in Australasia in 2011.

Points he made were:

- Capacity still exists but the previous view that Australasia offered a diversification benefit was under review;
- Price increases have already been seen and are expected to continue into the near future, particularly on loss affected layers of cover.

He also discussed some of the scepticism around catastrophe modelling in light of the earthquakes. Some of the areas of weakness in the models that have become apparent are:

- Underestimation of the liquefaction effect and the impact on land damage; and
- The difficulty in allowing for the impact of EQC cover.

Richard Dean from the Reserve Bank of New Zealand gave an update on the emerging prudential regulatory framework in New Zealand, and the challenges of introducing this during a period of turbulence, both seismically and economically.

And finally we had an update from Ross Simmonds and Jonathan Nicholls on the PwC Risk Margin Survey, outlining how NZ insurers' risk margins compare, and how they have been impacted by the release of the Actuaries Institute 2008 paper "A Framework for Assessing Risk Margins".

All in all an extremely interesting day at a challenging time for the industry. The presenters emphasised the numerous challenges faced by actuaries when dealing with the financial consequences of the earthquake; from improving catastrophe models to splitting insurance losses between parties. A constant refrain from nearly all presenters (including the scientists) was the key role actuaries can play in the collation, compilation and analysis of earthquake damage data.

NEW ZEALAND EARTHQUAKE DAMAGE, WARPED FOOTBRIDGE – © LOIS MCARTNEY | DREAMSTIME.COM

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NZSA Financial Services Forum 2011

The 2011 Financial Services Forum in November presented an intriguing concept: visually as well as verbally "Make sure you have your togs on!". This was a broad session and a number of presentations covered subjects generally not part of an actuary's daily life, very topical and relevant nonetheless, providing new and valuable insights.

There were a variety of presenters; from across the Tasman, Paul Newfield (Towers Watson) provided us an overview and update on Australia's current, and ever changing, post retirement environment. Closer to home; Richard Dean (Reserve Bank of New Zealand) provided an update on the implementation and development of the "Insurance Prudential Supervision Act – One Year on".

William Black (McGrath Nicol) shared with us his experience on "Governance and Corporate failures" including early warning signals that had been ignored. This was also a theme picked up by the following speaker, John Judge (ACC, ANZ) who as Board Chair for two of the largest NZ financial institutions', presented his view on "Board – Actuary relationships and what Boards want".

John Judge's view is that the Insurance Prudential Supervision Act (IPSA) and the increasing relevance attributed to risk management provide a significant opportunity for actuaries to strengthen their relationships with Boards. He provided clear views and guidance on what Boards want and need to enable them to make better informed decisions. He encouraged actuaries to support Boards in the following ways:

- to understand (technical) financial information better through training;
- to provide clear reports;
- to raise issues early and offer solutions, and
- to articulate disagreements with management.

Louis Boulanger (LB Now Ltd), as the last speaker of the day, presented his "Reflections from an Authorised Financial Advisor" in light of the recently strengthened NZ regulations for financial advisors. He was joined by Bradley Kidd (Chapman Trip) who presented the Church vs Armitage case in which a financial advisor, working as an employee, was held personally liable for losses incurred by the client despite the existence of a corporate 'veil'.

Both events were well organised and the presentations generally exceeded expectations. At the end of each day participants had gained valuable new insights and knowledge and as importantly had the opportunity to catch up with other members and participants. All in all the events were a great success.

Welcome new members – March 2012

NEW MEMBERS – AUSTRALIA

Ahmed, Mudassir	WA
Aliev, Roman	VIC
An, Kwanghyun	NSW
Bai, Annie	NSW
Bonarius, Nathan	NSW
Bong, Derwin	WA
Boss, Daniel	ACT
Chan, Matthew Ho Yan	VIC
Chen, Ruitong	NSW
Cheng, Kai Yuan	NSW
Cheung, Ka Ho	VIC
Chow, Raymond	NSW
Cullinane, Erinn	VIC
Fortmann, Werner	NSW
Freudenstein, Donald	NSW
Haggith, Harrison Rolf	NSW
He, Ting	NSW
Huang, Yixuan	NSW
Huang, Ying	NSW
Kiriella, Harshana Bandana	NSW
Ko, Thomas	NSW
Leung, Philip Lok Yin	VIC
Li, Zhe	NSW
Lin, Guang Xu	NSW
Liu, Yang	NSW
Liu, Lucy Cheng Cheng	VIC
Liu, Zhen	NSW
Looi, Leon Hua	VIC
Luo, Weldon Chuen Hung	NSW
Luy, Grace Pei-Yeng	NSW
Ng, Michelle	NSW
Niranjan, Nishan	NSW
Panchani, Tulsi Laxman	WA
QI ,Tianli	ACT
Qiu, Hao	VIC
Radhay, Neha	WA
Robinson, Matthew David	NSW
Sanghavi, Yash	WA
Schwalb, Richardt Emil	NSW
Simonson, Nathan Sydney	NSW
Siritharan, Vethooshan	WA
Sriramula, Mounica	NSW
Su, Lu Yi	NSW
Sun, Liwen	NSW
Sun, Chi	NSW
Sun, Puzheng	NSW
Tay, Andrew Kheng Kie	NSW
Teo, Joshua Khiang Teck	QLD
Tseng, Isaac Hsiang An	NSW
Wang, Songmin	ACT
Wang, Steven Jue	NSW
Wang, Zhiqi	VIC
Wani, Sagar Vivek	WA
Weenink, Monika	VIC
Wong, Kenneth Yuen Hui	NSW
Wong, Long San	NSW
Wu, Yi Kai	NSW
York, Matthew Gerard	VIC
Yuan, Yuan	NSW
Zhang, Lequn	WA

NEW MEMBERS – OVERSEAS

Chan, Yui Lam	Hong Kong
Chang, Wonsuk	Korea (S)
Choing, Clayton	Malaysia
Chiu, Hei Man	Hong Kong
Hou, Qian	New Zealand
Khor, Kok Chin	Malaysia
Liao, Weijin	Singapore
Lu, Ang	Singapore
Ng, Ka Chun	Hong Kong
Ong, Eu Chin	Malaysia
Primrose, Samuel James	New Zealand
Suen, Chi Hang	Hong Kong
Thaviwatanachaikul, Teerapong	Thailand
Watson, Emma Elizabeth	New Zealand
Zhang, Lingbo	China
Zhu, Yueyao	New Zealand

Congratulations to our 2011 prize winners

RESEARCH PRIZES

A M PARKER MEMORIAL PRIZE

Hugh Miller for his paper "Towards a Better Inflation Forecast" presented at the Actuaries Institute 2010 General Insurance Seminar

H M JACKSON MEMORIAL PRIZE

David Knox for his paper "The fairness and future of Australia's retirement income system" published in The Australian Economic Review, Vol 43, no. 3


Hugh Miller


David Knox

EDUCATION PRIZES

ANDREW PRESCOTT MEMORIAL PRIZES

Major Prize for best overall performance in the Fellowship examinations
Garreth Sweeney

Meritorious Performance 2011 Investments Examination
Yunting (Jacqueline) Liu

Meritorious Performance 2011 Life Insurance Examination
Jesse Haines

Meritorious Performance 2011 Global Retirement Income Systems Examination
Joel Atputharaj

Meritorious Performance 2011 Investment Management and Finance Examination
James Wu

Meritorious Performance 2011 Commercial Actuarial Practice Examination
Aniket Das

KATHERINE ROBERTSON MEMORIAL PRIZE 2011

Best performance 2011 General Insurance Examination
Andrew MacKessack


Garreth Sweeney


Yunting (Jacqueline) Lui


Jesse Haines


Joel Atputharaj


James Wu


Aniket Das


Andrew MacKessack

Actuaries Institute

Leading Experienced Practitioners and Academics Pathway to CERA

The Actuaries Institute is pleased to announce that it is now able to offer members the opportunity to be granted the Chartered Enterprise Risk Actuary (CERA) designation via the Leading Academics and Experienced Practitioner Pathway.

This is a marvellous one-off opportunity for those who have been practising in Enterprise Risk Management (ERM) to gain a formal, globally recognised qualification.

The CERA qualification is a formal recognition of risk management expertise founded on the principles of actuarial science and the framework of ERM. As described on the global CERA website (www.ceraglobal.org), the CERA Enterprise Risk Management credential is the most comprehensive and rigorous globally-recognised ERM designation and is supported by 13 member associations in 12 countries worldwide, including many of the major economies.

At the Actuaries Institute we already have 39 members who have achieved the CERA designation by completing an extensive ERM course (Course 7A) comprising the UK ST9 exam and a two-day workshop. However, the CERA qualification has now become much more accessible with the introduction of the one-off Leading Academics and Experienced Practitioner Pathway.

PROGRAM

The pathway program encompasses an independent panel of individuals with different experiences in risk management (both commercially and academically) and those who are familiar with the CERA qualification (in particular the requirements of the current course structure underlying the qualification). The role of the panel is to make recommendations in relation to all applications received during the application period, with final approval resting with Council.

Kent Griffin
Convenor, Risk Management Practice Committee
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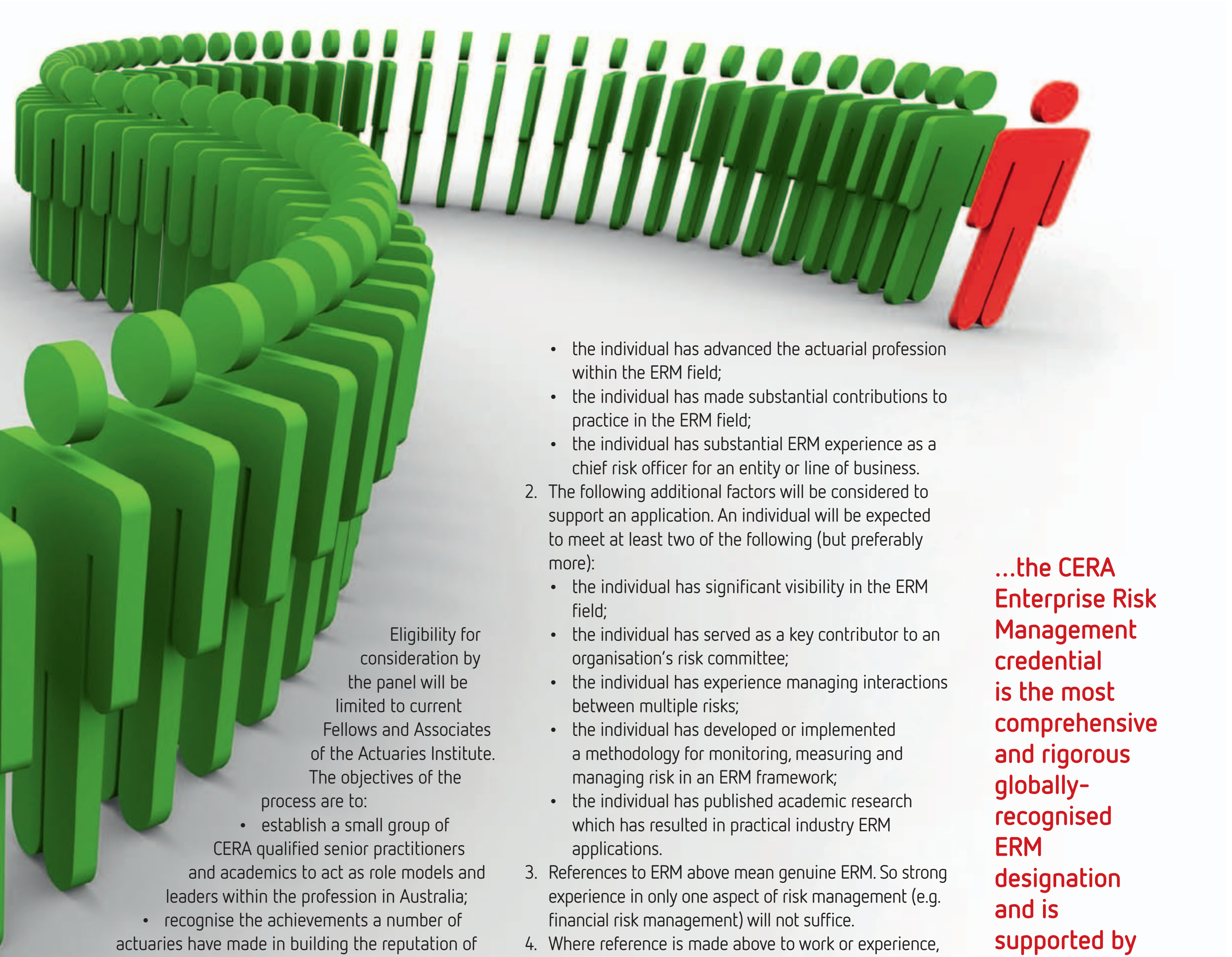

- the individual has advanced the actuarial profession within the ERM field;
- the individual has made substantial contributions to practice in the ERM field;
- the individual has substantial ERM experience as a chief risk officer for an entity or line of business.

2. The following additional factors will be considered to support an application. An individual will be expected to meet at least two of the following (but preferably more):
   - the individual has significant visibility in the ERM field;
   - the individual has served as a key contributor to an organisation's risk committee;
   - the individual has experience managing interactions between multiple risks;
   - the individual has developed or implemented a methodology for monitoring, measuring and managing risk in an ERM framework;
   - the individual has published academic research which has resulted in practical industry ERM applications.
3. References to ERM above mean genuine ERM. So strong experience in only one aspect of risk management (e.g. financial risk management) will not suffice.
4. Where reference is made above to work or experience, such work or experience must be substantial and must be for at least three years.
5. All applications are to be supported by written explanations as to why the criteria are met in the view of the applicant, with names of two referees who can support the applicant's submission.

Applicants who have been approved by Council for the CERA designation via LEPAP will also be required to complete the Institute's two day ERM workshop or have previously attended the Institute's five day ERM CPD course, which was offered from 2007 to 2009. A special separate session of the two day ERM workshop for LEPAP approved members will be held in early 2013.

Many members of the Actuaries Institute are already respected risk management practitioners, having worked and contributed in the field across a range of industries. This one-off process will offer an opportunity to formally recognise their extensive experience, and continue to build the reputation of our profession in this field. 🅰

Eligibility for consideration by the panel will be limited to current Fellows and Associates of the Actuaries Institute. The objectives of the process are to:

- establish a small group of CERA qualified senior practitioners and academics to act as role models and leaders within the profession in Australia;
- recognise the achievements a number of actuaries have made in building the reputation of the profession within the broader risk management communities of professionals, both academics and practitioners; and
- recognise the expertise, experience and research undertaken by actuaries in this area.

The process will be undertaken once only, and will likely have a six month period during which applications will be taken.

HOW CAN I APPLY FOR CERA VIA THE PATHWAY?

A communication was recently sent to all members calling for applications.:

1. Applicants should have a minimum of three years of substantial ERM experience and should meet at least two of the following criteria (but preferably three or four):
   - the individual has performed work in the field of ERM at a senior level;

...the CERA Enterprise Risk Management credential is the most comprehensive and rigorous globally-recognised ERM designation and is supported by 13 member associations in 12 countries worldwide, including many of the major economies.

Welcome to "Staying Ahead"

The Institute's 2012 Business Plan includes a project on re-vamping CPD, with new initiatives and improvements.

This new monthly column will be your source for Continuing Professional Development (CPD) information. You will be able to find out about what the Institute is planning, new initiatives that are being launched, how they will improve your current CPD experience, and how you can get involved.

The Institute's 2012 Business Plan includes a project on re-vamping CPD. The project objective is to enable actuaries in traditional fields to efficiently and effectively source professional development experiences to support, maintain and grow their capability in their roles as well as be relevant to actuaries working in broader fields.

We have conducted an extensive review of the current CPD Program and plan to make a number of changes including:

- enhanced functionality for members to record and monitor their CPD;
- capability mapping to CPD offerings;
- implementation of new CPD offerings; and
- increased networking opportunities.

Work undertaken in other Business Plan projects will provide both the tools and technology to support the achievement of the CPD re-vamp project outcome.

FELLOWS USING PART III AS CPD – AN INTERVIEW

Last September, the Institute announced that Fellows could register for access to Part III course sites in the Learning Management System (LMS)for CPD purposes (free of charge).

Part III courses in the LMS contain a news forum, course materials, course readings, tutorial presentations, exercises and audio recordings, and discussion forums for the current semester, as well as learning assets from past courses for tutorials, exam questions and solutions, assignment questions and solutions and discussion forums.

Since this announcement, 26 Fellows have accessed the course site in the LMS. We interviewed some of these Fellows to find out about their experience.

Thank you to all who responded. We would encourage Fellows, who have not yet applied to access the Part III courses in the LMS, to consider it as part of their 2012 CPD planning.

What courses did you choose and why?	Respondents said that they chose all courses because they; • wanted the materials to be handy as a resource to refer to; • read the forum posts to understand current issues in areas where they had only basic knowledge; and • read up on practice areas they had an interest in.
What did you experience and learn?	Respondents learned about practice area current issues and hot topics. Some stated that their original idea of choosing all was bigger than the appetite they had to look deeply into the courses.
How can you use what you have learned in the future?	Respondents stated that they will use the courses to learn about elements where they feel they are weak, as it can be difficult to keep up with all developments, so having access to the course will be good to close that gap. Other respondents stated that they have enhanced their interest level in areas which will likely lead to serious self study in areas relevant to their current work.
How much time have you spent accessing courses?	Responses varied from: • Not yet. • When I need to get the information. • 15-20 min each day checking forum posts.
Do you recommend Part III CPD to other Fellows? Why? Why not?	Respondents all said YES, reasons included:: • Useful to keep up with current developments in other practice areas; • Good resource if you are looking to enter a new practice area; • it's a free resource, value for membership so, "why not?"

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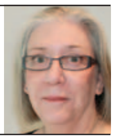

CPD SURVEY – WHAT'S IN A NAME?

Thank you to all who responded to our survey, CPD – What's in a Name. Key points from the responses included:

- We should keep the term CPD, as it is used and known in organisations worldwide.
- Several ideas were put forward to find a new set of words to fit CPD including;
  - Constantly Developing with Purpose.
  - Career Promotion Driver.
  - Some others we could not possibly use, but we did enjoy the humour…

Many ideas centered around improvement and the future. After much discussion and consultation agreement was reached that, the new "tag line" for the Institute's CPD will be, *Staying Ahead*.

Staying Ahead refers to the opportunities that CPD presents to members, enabling them to always remain current and ahead in their work practices and careers.

The new tag will be represented as;
CPD – *Staying Ahead*.


Ros Calver

Congratulations to Ros Calver from Towers Watson who suggested *Staying Ahead*. A token of the Institute's appreciation has been sent to Ros in recognition of this innovative suggestion.

WHAT CPD INITIATIVES WILL YOU HEAR MORE ABOUT IN THE NEXT ISSUE?

CPD Member Surveys were sent out at the end of 2011 and beginning of 2012. The survey results have been collated and are being presented to Practice Area Committees to assist them in planning for their 2012 events calendar. In the next issue, hear how the data collected in the survey is being used to meet member needs.

Think CAPABILITY…it's coming!

Notices

Disciplinary Notices

Re GRAHAM PETER OSBORN

On 19 May 2011, a Complaint was made under the Disciplinary Scheme of the Institute of Actuaries of Australia by the Australian Prudential Regulation Authority against Mr Graham Peter Osborn.

The Complaint arose from an Insurance Liability Valuation Report as at 30 June 2008 prepared by Mr Osborn in relation to ACN 000 007 492 Limited (formerly Rural & General Insurance Limited) (In Liquidation), specifically that Mr Osborn had failed to comply with APRA's General Prudential Standard 310 (Audit and Actuarial Reporting and Valuation) and with the Institute's Professional Standard 300 in preparing that report.

On 21 December 2011, Investigating Sub-Committee 1004 determined that there was a *prima facie* case of Actionable Conduct (as defined in Rule 3.2(a)(ii) of the Scheme, namely 'unsatisfactory professional conduct') having been committed by Mr Osborn in failing to comply with those standards.

The Investigating Sub-Committee's report was referred to the Convenor of the Tribunal Panel who has formed Tribunal 12.01 to hear and determine the matter in accordance with the Disciplinary Scheme.

6 March 2012

Re ANTHONY RICHARD LEWIS

On 19 July 2010, a Complaint was made under the Disciplinary Scheme of the Institute of Actuaries of Australia by Ms Melinda Howes (in her capacity as Chief Executive Officer of the Institute) against Mr Anthony Richard Lewis.

The Complaint arose from the Respondent having been made 'insolvent under administration' (which constitutes *prima facie* evidence of Actionable Conduct pursuant to Rule 3.2(d)(v) of the Disciplinary Scheme) and failing to so advise the Institute (in breach of Rule 3.3 of the Disciplinary Scheme). The Complaint alleged that Mr Lewis' conduct was likely to bring discredit upon the Institute or the profession of Actuary and, therefore, constituted "Actionable Conduct" under Rule 3.2(a)(iii) of the Disciplinary Scheme.

In the course of the investigation by Investigating Sub-Committee 1003 ("ISC 1003"), other matters arose concerning Mr Lewis' conduct and which suggested that he had engaged in Actionable Conduct other than as strictly set out on the face of the Complaint. On 30 November 2010, these other matters were formally referred to ISC 1003 by the Convenor of the Professional Conduct Committee for further investigation.

On 6 February 2012, ISC 1003 determined that there was a *prima facie* case of Actionable Conduct having been committed by Mr Lewis.

The Investigating Sub-Committee's report was referred to the Convenor of the Tribunal Panel who has formed Tribunal 12.02 to hear and determine the matter in accordance with the Disciplinary Scheme.

6 March 2012

Philip Latham
Head of Education

Education Update – March 2012

PART III RESULTS SEMESTER 2 2011

Semester 2 2011 saw the Part III pass rate drop to 33% (including non-Fellows only for C7A ERM), which is the lowest it has been since Semester 1 2005, where the pass rate was 30%. The Part III pass rate excluding the C7A ERM course was 33 % in Semester 2 2011 and 42% in Semester 1 2011. The pass rates since 2006 (including C7A for non-Fellows since 2010) are as follows:

2011 (2)	2011 (1)	2010 (2)	2010 (1)	2009 (2)	2009 (1)	2008 (2)	2008 (1)	2007 (2)	2007 (1)
33%	36%	40%	40%	40%	44%	49%	44%	41%	38%

Low pass rates in the C7A ERM subject have contributed to lower overall Part III pass rates in recent semesters. However, in Semester 2 2011, very low pass rates in both Life Insurance courses were the biggest contributor to the low overall pass rate. The possible causes of the low pass rates in these subjects and measures to address it have been discussed by the Education Council Committee.

Below are the pass rates for each course in semester 2 2011 compared with the last three semesters:

	2011 (2)	2011 (1)	2010 (2)	2010 (1)
Course 1 Investments	31%	33%	31%	35%
Course 2A Life Insurance	20%	30%	31%	28%
Course 2B Life Insurance	15%	39%	41%	44%
Course 3A General Insurance	23%	33%	36%	37%
Course 3B General Insurance	31%	34%	40%	35%
Course 5A Investment Management and Finance	62%	n/a	53%	n/a
Course 5B Investment Management and Finance	n/a	38%	n/a	56%
Course 6A Global Retirement Income Systems	n/a	50%	n/a	25%
Course 6B: Global Retirement Income Systems	63%	n/a	54%	n/a
Course 7A Enterprise Risk Management*	26%	21%	34%	19%
Course 10 Commercial Actuarial Practice	55%	59%	55%	59%

* pass rates for C7A ERM above are for non-Fellows only. The pass rates for Fellows in the C7A ERM subject have been as follows:
Semester 1 2010: 10 sat, 6 passed, (60%). Pass rate for all members (25%)
Semester 2 2010 10 sat, 8 passed, (80%). Pass rate for all members (40%)
Semester 1 2011: 3 sat, 0 passed, (0%). Pass rate for all members (20%)
Semester 2 2011: 7 sat, 2 passed, (29%). Pass rate for all members (26%)

PASS RATES AND EXAM CENTRES

The pass rates by exam centre for the last four semesters is as follows:

	2011 (2)	2011 (1)	2010 (2)	2010 (1)
Sydney	35	35	42	39
Melbourne	36	40	45	57
Other Australian	24	52	26	40
Overseas	21	33	31	37

Pass rates for overseas exam centres and Australian exam centres other than Sydney and Melbourne had been improving over the years, but dropped significantly on the previous semester.

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"All the leaves are brown, but the sky isn't grey..."

REFLECTIONS FROM THE 2011 UNIVERSITY OF RIVERSIDE GLOBAL CONFERENCE ON EDUCATION

Top to bottom: a quiet Thursday morning at Downtown Riverside; the author presenting; some of the delegates stand together for a group photo at the close of the conference

I recently presented at the 2011 University of Riverside Global Conference on Education which ran from 18-19 November 2011 at the Hampton Inn, Corona East, California in USA. On top of this, I decided to bolt on a leisure trip , arriving at the USA two days before the conference. All in all, this trip was most memorable and I would be inclined to write all about it here, but let's leave it for another article…

Now about the conference!

The conference runs twice a year by the School of Education at University of Riverside, bringing together delegates from around the world to present the latest research findings in education methodology and policy setting. Delegates came from disciplines ranging from sociology, psychology to medical sciences. Oddly enough, amongst the thirty odd delegates I was one of three from the commerce discipline.

Corona East is in Riverside County, 80km from Downtown Los Angeles. What struck a chord with me as I first arrived there was the amazingly pleasant weather given it was late fall. Arriving at my hotel though was no easy feat given I touched down in LA. I had resisted sleeping for 30 hours to fight jet lag and had to take public transport to get there. For those not aware of Los Angeles, public transport there makes Sydney's system look state-of-the-art!

The conference started off with an opening address by Dr Dan Smith, also known globally as Pastor Dan. His address doubled up as a sermon praising the value of education and the impact it has on shaping societal values. I personally found this inspiring, as this has been my personal motivation to embark on a career in higher education.

Delegates presenting covered a diverse range of topics. Notable presentations included the shift towards online schools in the US from traditional schools, programs to develop multicultural awareness in residential colleges, state government subsidies to promote higher education in India and the development of better school-parent partnership for primary school students coming from Hispanic backgrounds.

The paper I presented – "What perceptions do actuarial graduates hold on their university studies, work experience and career direction after completing the university program", is based on a survey study conducted by Associate Professor John Evans and Dr David Morgan from the University of New South Wales (UNSW). This study sought the views of actuarial studies graduates who recently completed their degree program at UNSW.

Key findings from this study include:

- Respondents were satisfied about pursuing actuarial studies as their career path;
- The actuarial qualification process is time consuming with Part III being particularly demanding though moderately relevant to their job;
- Career advancement potential and employer reputation ranked as being most important in impacting on the respondents' decision in selecting their first job after completing university studies;
- Salary became increasingly important three to five years after leaving university;
- Longer term career aspirations include being in senior management and research; and
- Programs on development of generalist skills necessary for management roles are only moderately enticing to the respondents.

I received some useful feedback from the delegates, the general consensus is for a review on the actuarial education program to identify the core principles and material. They recommended not letting our education program shift to accommodate the commercial world as we risk losing the foundational knowledge which defines and distinguishes our actuarial science discipline.

Whilst this conference did not offer a gala dinner a delegate invited me to join his family for a Vietnamese style dinner on the first night. On the second night Dr Raj Singh, also President and CEO of University of Riverside, invited another delegate and myself to join his family to an Indian dinner and a night out at Anaheim. We went to Disneyland that night, though we did not go inside the theme park. Nevertheless, there's so much to do outside the theme park with all the stores and street performers!

Valuable networks were established and since the close of the conference, I have kept in touch with a number of the delegates. One of the delegates has since rallied together a number of delegates and begun seeking collaborators for his research ventures. I would like to thank Associate Professor John Evans who encouraged me to present at this conference and reviewed the paper. I also thank Lesley Traverso from DW Simpson Asia-Pacific for her ideas which led to this survey. Finally, I extend my thanks also to Dr Adam Butt and Associate Professor Anthony Asher for their input. A

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Australian National University **hosts 4th Australian Actuarial Education and Research Symposium**

Benjamin Avanzi, UNSW

33 presentations on a wide variety of actuarial topics showed that actuarial research is alive and well in Australia.

The School of Finance, Actuarial Studies and Applied Statistics at the ANU in Canberra was proud to host the largest Actuarial Education and Research Symposium to date on 1 and 2 December 2011.

How large might that be you might ask: well you may be surprised to learn that more than 60 attendees represented all six universities that teach actuarial studies in Australia, with overseas delegates from New Zealand, Switzerland and Spain and many other interested parties.

33 presentations on a wide variety of actuarial topics showed that actuarial research is alive and well in Australia. Presentations from all the universities and other contributors were made by PhD candidates, recent Honours graduates and university staff. Themes ranged from mortality and disability income insurance, portfolio selection and general insurance to the use of household microsimulations, measures of subjective financial well-being and modelling postcode level house price risk.

Education received a special focus. A set of education related concurrent sessions was topped off with a plenary session followed by a lively panel discussion facilitated by John Shepherd. The panel discussion addressed current issues and the changing landscape for the actuarial education process in Australia. We were fortunate to have as panellists Trever McMahon,current Chair of the Education Council Committee, Philip Latham, Education Manager at the Actuaries Institute and David Service, Course Leader of the Commercial Actuarial Practice course and convenor of the Actuarial Control Cycle courses at ANU.

The event was actively supported by the Actuaries Institute with opening comments from the President Barry Rafe and contributions by the Chair of the Research Council Committee, Chris Latham and the editor of the Australian Actuarial Journal, Andrew Leung.

This level of interaction between the universities themselves and between the universities and others with a vested interest in actuarial education and research can only help to improve the effectiveness of all our efforts in these domains. If you are interested in more detail on the programme and presentations, please contact Tim Higgins (tim.higgins@anu.edu.au) or the author. If you have an interest in the education process in general you might like to take a look at the Actuarial Educators Network at http://www.actuaries.org/AEN/

The next symposium will be at Monash University in Melbourne late in 2012: we'd love to see you there!

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Creating time and space to think

Does it seem to you that time is accelerating? It's not just that I'm getting older (although that's probably part of it). It's also that as each year goes by we have more and more demands on us in our day to day jobs. More channels of communication, a higher volume of information, less time to assimilate that information and make decisions.

Many people have to field more than 100 emails a day as well as phone calls, text messages, tweets, LinkedIn or Facebook messages… and that's before we start on our "To Do" list of actual jobs.

Then it seems that there is always some emergency. Something that needs to be done right now and cannot wait, or something that needs to be signed off for someone else…the list goes on.

Many of us are in back-to-back meetings from 9am until 5pm each day, and then we start the day's work after that.

And more is expected of us as well. We aren't just expected to be technicians; we are expected to be creative and innovative. We are supposed to think strategically. We need our radar out for what's happening in our industry and beyond.

If we manage people, it's not good enough anymore to just delegate work to them and set clear expectations. We need to involve them in decision making, to empower them and get their buy-in. We are expected to be sensitive to the needs of our team and take time out to listen to them and to make sure we give them lots of positive feedback. And above all, we need to be available so they can have a quick chat when needed.

With so many demands for our attention it is harder and harder to spend time on the things that matter – those non-urgent but very important things that actually help us achieve our business goals, complete our projects and work on strategic initiatives. It's also a challenge to find the time we need to put in to be a leader of people – to coach and mentor our team to allow them reach their full potential.

These are all things that are a real challenge for me, and I am not alone. Many of my friends and colleagues, especially those juggling people leadership responsibilities, are in this situation too. You start to feel like you are drowning – that no matter how much you work there will always be more work and there is nothing you can do to get out from under it.

What can be done? Is there hope?

ATTENTION DEFICIT TRAIT: FIRE-FIGHTING MODE AND WHAT IT DOES TO OUR BRAIN

I recently read an interesting article recently by Katharine McLennan* which applies insights from neurosciences to questions of leadership and culture in organisations. Katherine refers to an article in January 2005, in the Harvard Business Review called "Overloaded Circuits: Why Smart People Underperform." It describes an executive going through some of the problems I outlined above – I think of this as operating in fire-fighting mode. I quote from Katharine's article:

> **"Unlike attention deficit disorder (ADD)—a neurological disorder that has a genetic component and can be aggravated by environmental and physical factors—attention deficit trait (ADT) springs entirely from the environment. Never in history has our brain been asked to track so many data points. When the mind is coping, it is being governed effectively by the frontal and prefrontal lobes of the cortex, which guide our decision making and planning, the organisation and prioritisation of information and ideas, and time management. When the mind is coping, the deep centres below the frontal lobes that govern basic functions like sleep, hunger, sexual desire, breathing and heart rate are sending up messages of satisfaction and joy. They are pumping up your attention and motivation and won't interfere with your working memory, which is what you need to track the many data points coming in.**
>
> **But when the brain suddenly has to deal with the sixth decision after the fifth interruption in the midst of the search for the ninth missing piece of information on the same day that the third deal has collapsed, the brain begins to panic, reacting as if it were responding to a sabre-toothed tiger attack. The deep centres now interpret the messages from the frontal lobes by sending alarm signals of fear, panic, anxiety and irritability. The frontal lobes are hijacked by these deep centres' messages and fail to assert their calm, rational decision making. We respond to challenging stimulation with anger, fear and anxiety. We are robbed of our flexibility, our sense of humour and our ability to deal with the unknown. We forget the big picture and the goals and values for which we stand. We lose our creativity and our ability to shift course. We fail to see the choice that humans alone as a species have to react to any situation… we ALWAYS have choice about how we can respond. We do not have to succumb to the hijack that wants to bypass our rational frontal lobes—IF our minds are in a healthy state capable of pressing that PAUSE button for even just a split second."**

Sound familiar?

SLOW YOUR BRAIN, AND LOOK YOUNGER!

It's vital that we can step away from this environment and create time and space to think. I discovered this by accident when I started my commuting lifestyle. I have 1-2 days a week where I work from my idyllic home in far northern NSW. This is my time to think, and I have found it invaluable. This is when I get my best ideas – I am creative. And it's because I slow down.

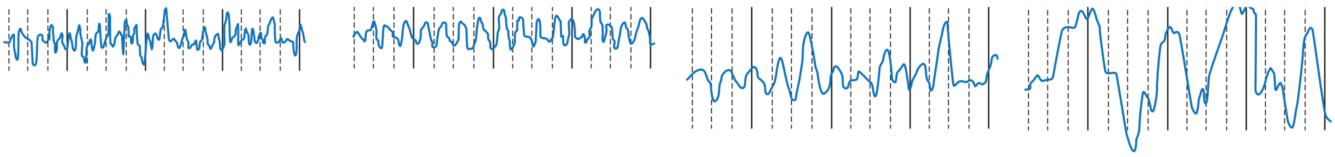
Katherine talks about a range of ways this can be done within the work environment. But her most interesting, and "out there" concept, was meditation.

There are four types of brainwaves, as the figure (see page 36) from the article shows.

Modern work environments with their multiple demands push us into fire fighting mode –encouraging us to produce beta brainwaves. This not only stresses us out, it also produces chemicals which age our bodies! But slowing our brainwaves – even just from beta to alpha – lets us access our creativity. Slowing further gets us to theta, where "out of the blue" ideas happen. Beneficial chemicals are produced which have positive impacts on ageing and increase our resistance to disease. Talk about a virtuous circle – we feel less stressed, get sick less and look younger. I want what she's having!

Meditation is one proven way to slow the brain. But there are others

FIGURE: FOUR CATEGORIES OF BRAIN WAVE PATTERNS



Beta (14-30Hz)
- Concentration, arousal, alertness, cognition
- Higher levels associated with anxiety, disease, feelings of separation, fight or flight

Alpha (8-13.9Hz)
- Relaxation, super learning, relaxed focus, light trance, increased serotonin production
- Pre-sleep, pre-waking drowsiness, meditation, beginning of access to unconscious mind

Theta (4-7.9Hz)
- Dreaming sleep (ERM sleep)
- Increased production of catecholamines (vital for learning and memory), increased creativity
- Integrative emotional experiences, potential change in behaviour

Delta (1-3.9Hz)
- Dreamless sleep
- Human growth hormone released
- Deep trance-like state, loss of body awareness
- Access to unconscious and 'collective unconscious' mind

– gardening, yoga and other forms of exercise are also beneficial. I guess it's a bit hard to garden in the office, but we have started doing a group yoga session two mornings a week in the secretariat office and I find I am more relaxed and productive on those days. A number of my team work a day or two a week from home, and report that they get through a lot of work and get "back under control" on those days. We are also making a conscious effort to reduce the number of meetings we hold – but that is an area where I need to do more.

We need to change the way we work, and the way we set up our teams to work. Maybe the above measures won't work in your office. But I am sure there are things you can do – the article has a range of suggestions and I commend it to you (full details in footnote below).

FOCUSING ON Q2

OK, so let's assume we are meditating or gardening or swimming regularly, we have found time and space to slow our brain which lets us regain control in situations of stress and to tap into our creative mind. (Big assumption…)

Now what are we going to focus on whilst we are not in all those meetings and responding to our 120 emails a day?

Dave Miller**, who gave those talks last year on The Influential Actuary CPD Tour, has been emailing regular leadership tips to some of us who opted in to his mailing list. In his latest email (Leadership Tip #16) he says:

> **"One thing that is totally essential if you want to be an influential leader is that you must master how you spend your time… ultimately how you manage your time comes down to how you live out your priorities.**
>
> **The Question: "What's one thing you could do (you aren't doing now) that if you did on a regular basis, would make a tremendous positive difference in your professional life, business or career?"**
>
> **Your answer will most likely fall into what Stephen Covey has termed "Quadrant II" – things that are Important and Not Urgent."**

Dave goes on to say:

> **"…the more time you can spend in Q II, the more successful you will be. Covey found this to be true. In his classic book The 7 Habits of Highly Effective People, Covey discovered that the average person spends less than 5% of their time in QII, while the most effective leaders spend at least 50% of their time there!**
>
> **Q II is the place of proactivity – these things require initiative from us… :**
>
> - **Strategic planning**
> - **Building relationships**
> - **Making follow up calls**
> - **Saying "no" to the urgent activities that distract me from my purpose**
> - **Finishing that proposal I've been procrastinating on**
> - **Exercise**
> - **Connecting with my vision every day**
> - **Following through on goals that will help my career on a daily basis**
> - **Honing my influencing skills"**

When you spend the day answering emails and dealing with urgent but unimportant tasks, you feel effective, busy and important. But you are not working effectively. I have never seen a set of goals or KPIs that included "answer every email on the same day". Most of us are measured on things like delivery of projects and key initiatives, sales, or your implementation of a strategy. I know from experience how hard it is to make yourself shut down Outlook and work on a strategy paper or write a media article whilst the emails are mounting up. But that's what being in QII is all about – creating the space and time to think – and then to ensure we spend that time on what's important.

LEADERSHIP A BIG FOCUS IN 2012

At the Actuaries Institute we have a real focus on developing actuaries as leaders. Your Leadership Committee under convenor Jenny Lyon will be rolling out a series of initiatives this year targeting members at all levels, from those training to be actuaries right through to senior people on boards.

Katharine McLennan, who wrote the neuroscience article I mentioned above, will present a concurrent session at the Financial Services Forum, being held in Melbourne on 30 April and 1 May. We have a great program which is all about 'Thinking Big' – the first step of leadership. I hope to see you there. A

* The Neuroscience of Leadership and Culture by Katharine McLennan, Mettle Group, March 2007

**You can contact Dave Miller at Dave@LeadershipGrowthStrategies.com

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Email: ActuariesMag@actuaries.asn.au for details

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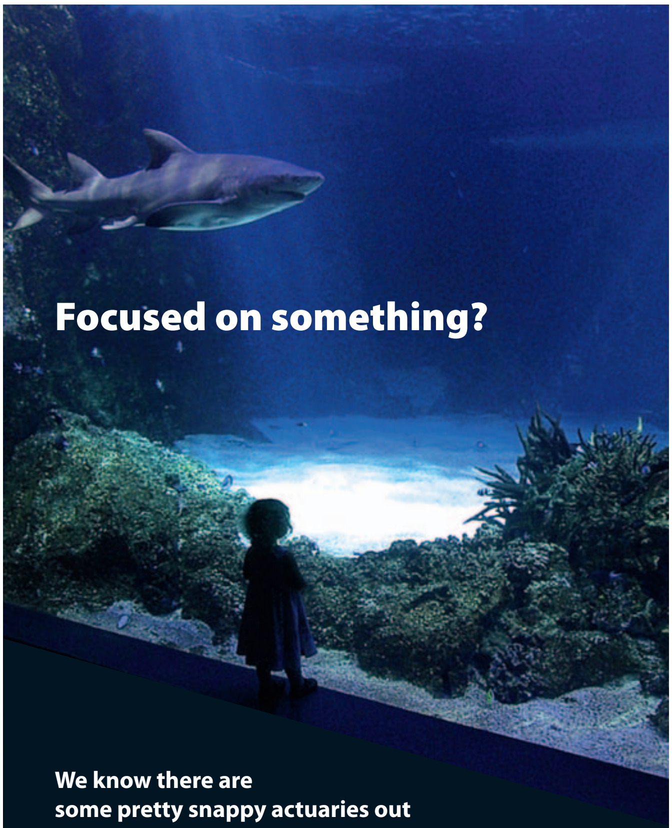
Meet Josephine.

Jo's just got a new job that her new company wants you to know about. Or perhaps she's moved but needs all her non social-media friends to know where to contact her. It could be she's ended a long-term relationship and has a slightly-used beemer she needs to divest… quickly. Or some jewellery. Or a hat collection. Maybe she's just written her first crime novel 'He'll get nothing!' that she wants to publicise. Whatever the reason, small space advertising like this one is now available in Actuaries magazine. At $300 it's possibly cheaper and more memorable than a night on the town.

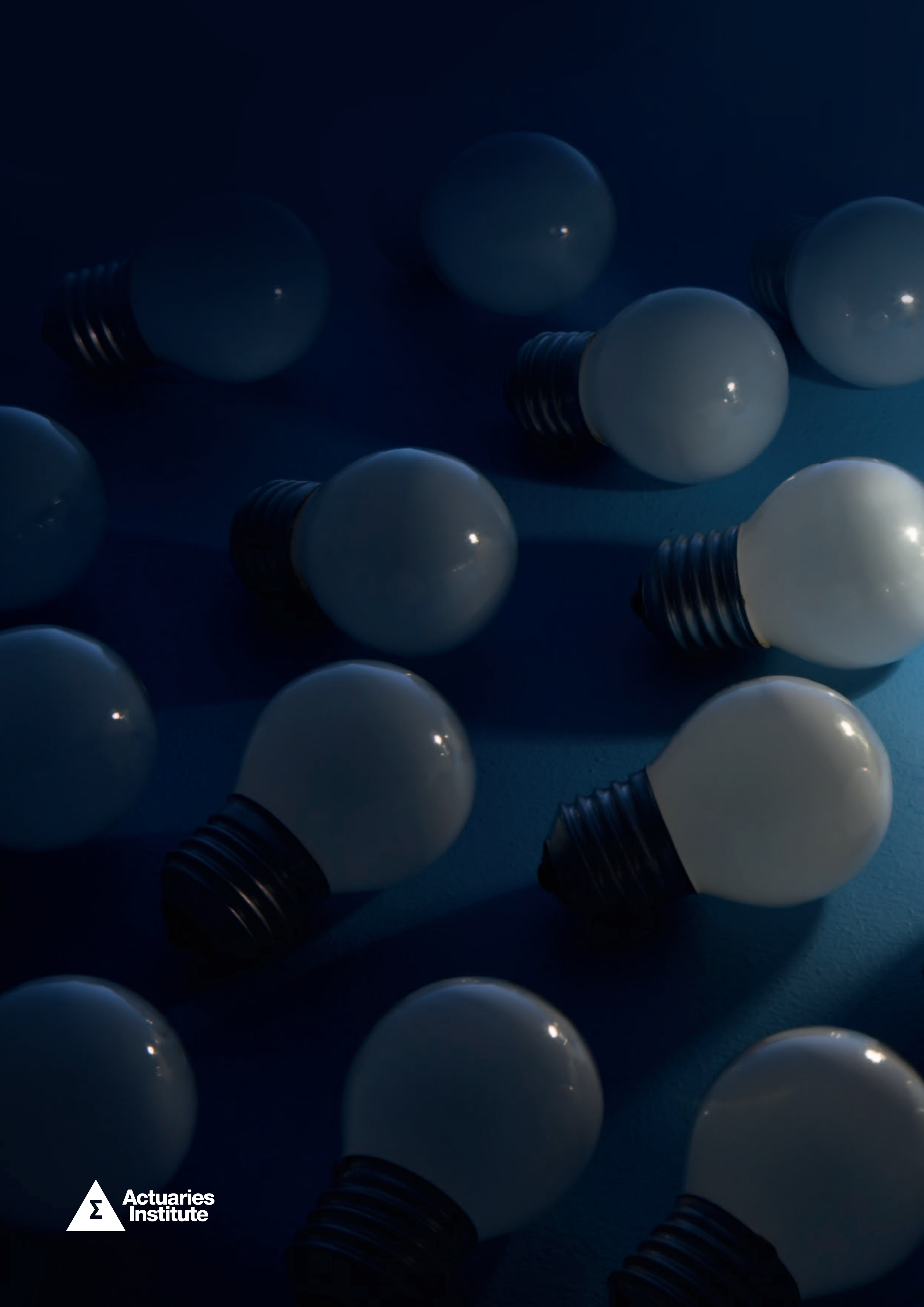


Focused on something?

We know there are some pretty snappy actuaries out there… so we're running a competition to find the Institute's best photographers. The winning photo will be reproduced on the cover of a future issue of Actuaries magazine. Fame awaits! (Fortune, maybe not). So see below for conditions or visit the website www.actuaries.asn.au for details.

Entry conditions

1. Submission must be made by a current member of the Actuaries Institute.
2. Theme: "Must have actuarial content". You can interpret this any way you want. Be creative! Be artistic! Be involved!
3. Images must be high resolution (200dpi min), JPEG format only (with your surname as part of the file name). Images size: minimum 5mb; maximum 10mb. If over 7mb, send multiple files separately. Incorrectly formatted, and insufficient resolution images will not be considered.
4. Closing date 5pm, 31 July 2012.
5. Results will be published later in the year.
6. The judging panel will consist of the Actuaries editorial committee. No submissions will be entered into, and decisions by the committee will be final.
7. The winning photograph will be published on the front cover of Actuaries magazine.
7. The editorial committee may, at their discretion award additional prizes or commendations.
9. Limit of 3 images per person.
10. The photographs must be your own.
11. If the photographs contain images of other persons, please ensure that they have provided authorisation and permission. Images that require the permission of third parties for commercial reproduction (such as the traditional landowners of Uluru) must be accompanied by a copy of the signed agreement.
12. Please provide your name and contact details (they will be subject to the Institute's standard Privacy Policy).
13. Copyright remains with the photographer.
14. By entering the copyright owner agrees to the image to be reproduced in the Actuaries magazine (with acknowledgement of the photographer) without charge.
15. You may provide a caption with the photograph. Captions are to be no greater than 20 words.
16. Email your image(s) to: ActuariesMag@actuaries.asn.au



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